

Forestry Impacts on BC Pacific Salmon

Musgamagw Dzawada'enuxw territory watersheds

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Introduction

Logging has impacted many rivers in BC that support salmon. Using historical data (1950-2012) of pink salmon abundance in 417 rivers, chum salmon abundance in 374 rivers in BC, and of logging disturbance in these rivers, we estimated the impact that logging has on salmon productivity (number of salmon produced in the next generation per spawner) using a statistical model. The two metrics of logging disturbance that we used are:

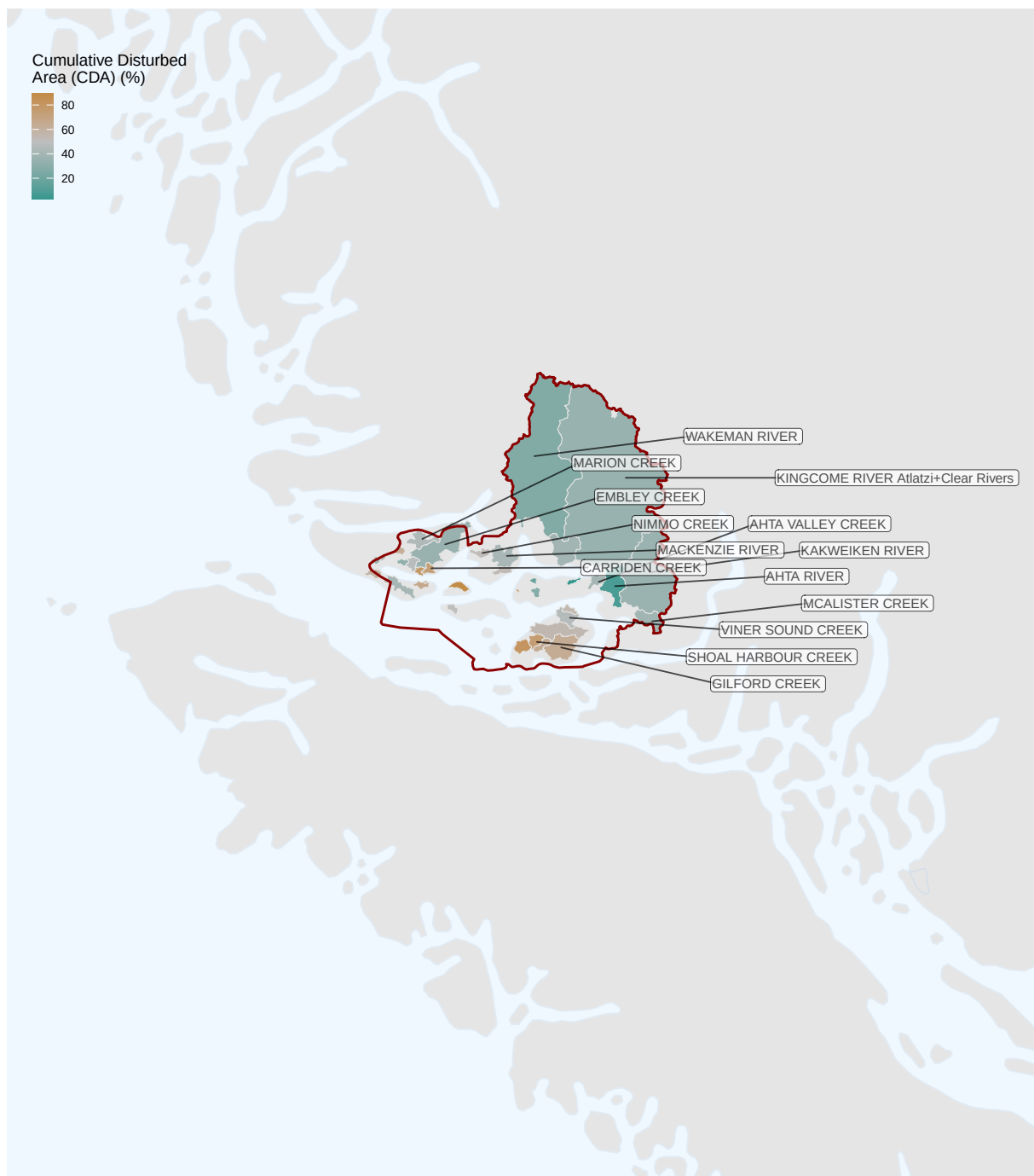
- 1) Cumulative Disturbed Area (CDA) - the percent of area around a river that has been disturbed by human activities or natural disturbances without assuming any recovery within the time period of our dataset (62 years).
- 2) Equivalent Clearcut Area (ECA) - proportion of area that has been clearcut around a river after accounting for an expected hydrologic recovery. This metric assumes that the forest is close to 100% recovered roughly 50 years post-logging.

In addition to logging disturbance, we also used two variables to account for ocean conditions:

- 3) Sea-Surface Temperature (SST) - the average sea-surface temperature corresponding to the time when the salmon first enters the ocean, which is known to have variable effects on salmon productivity.
- 4) North Pacific Gyre Oscillation (NPGO) - index of primary productivity in the ocean, which is known to have a positive effect on salmon productivity.

In this document, we have used the 4 coastwide models (with CDA for chum, with ECA for chum, with CDA for pink, and with ECA for pink) to estimate the impact of logging on some of the rivers within MD territory for which we had data. For each of these rivers, we have plotted the data of logging disturbance as given by the two metrics, ECA and CDA, and the number of salmon recruits with time. We then plotted the results of the model for that river A) the probability distributions of the effect of CDA, NPGO, and SST on salmon productivity for that river (probability distributions with the peak to the left of 0 indicate a negative effect on salmon productivity), B) the predicted spawner-recruit relationship for two different levels of logging disturbance experienced by the river, C) the predicted change in productivity for different theoretical levels of logging disturbance using CDA, and D) using ECA. For figure C, the dashed line corresponds to the value of CDA from 2022 experienced by the river. For figure D, the dashed line corresponds to the highest value of ECA (until 2022) experienced by the river.

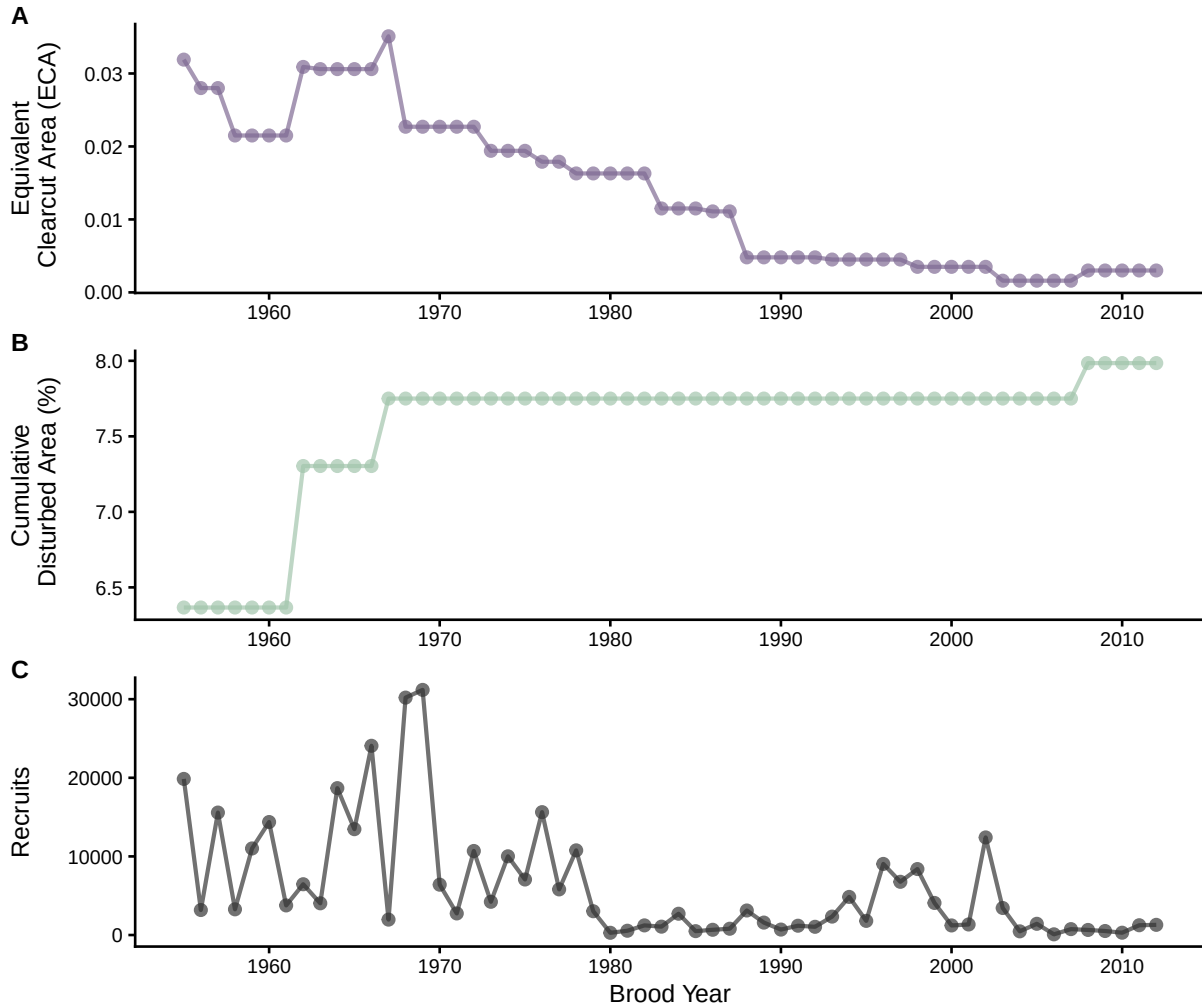
Map of MD territory watersheds



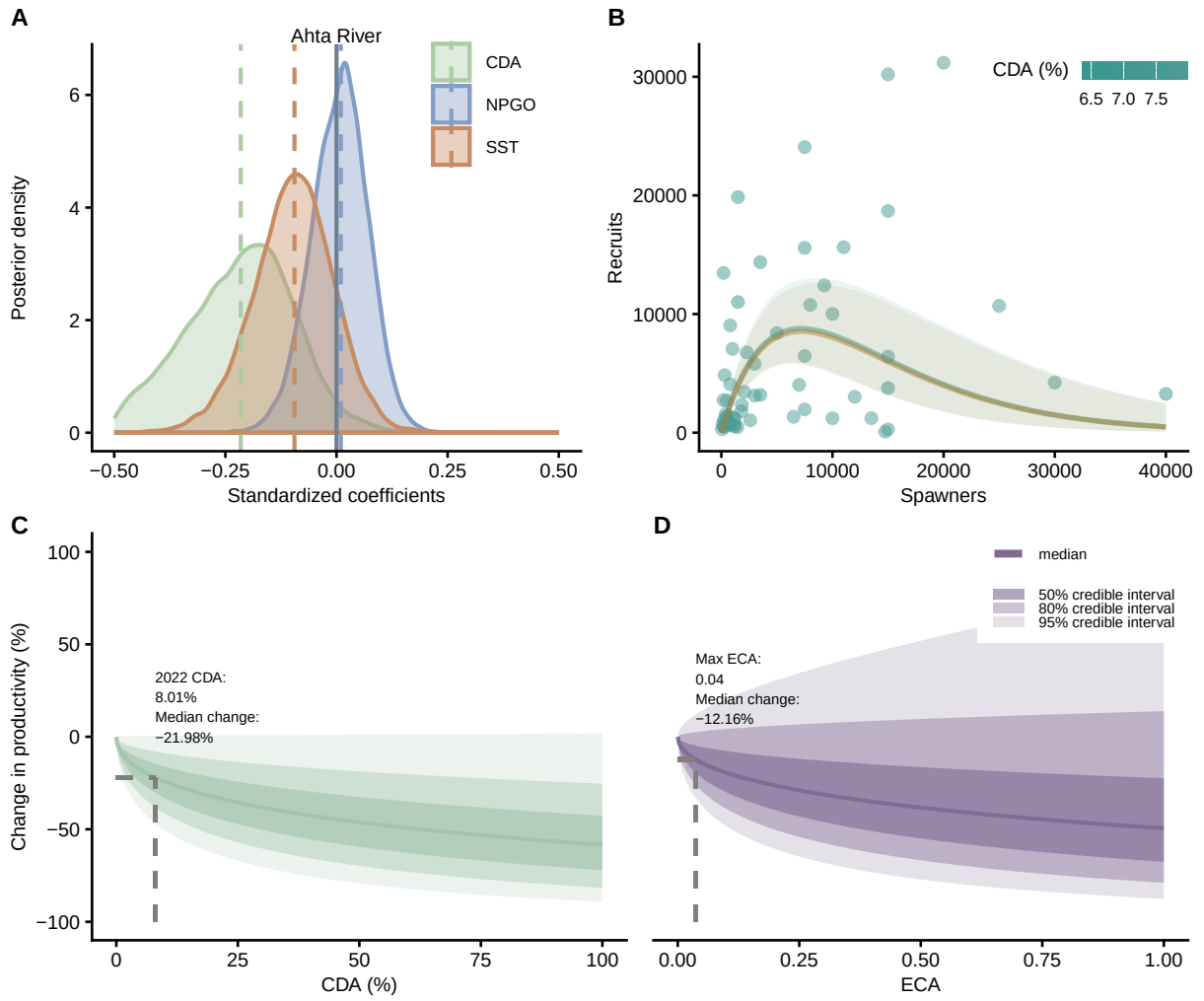
River-level data and model results

AHTA RIVER

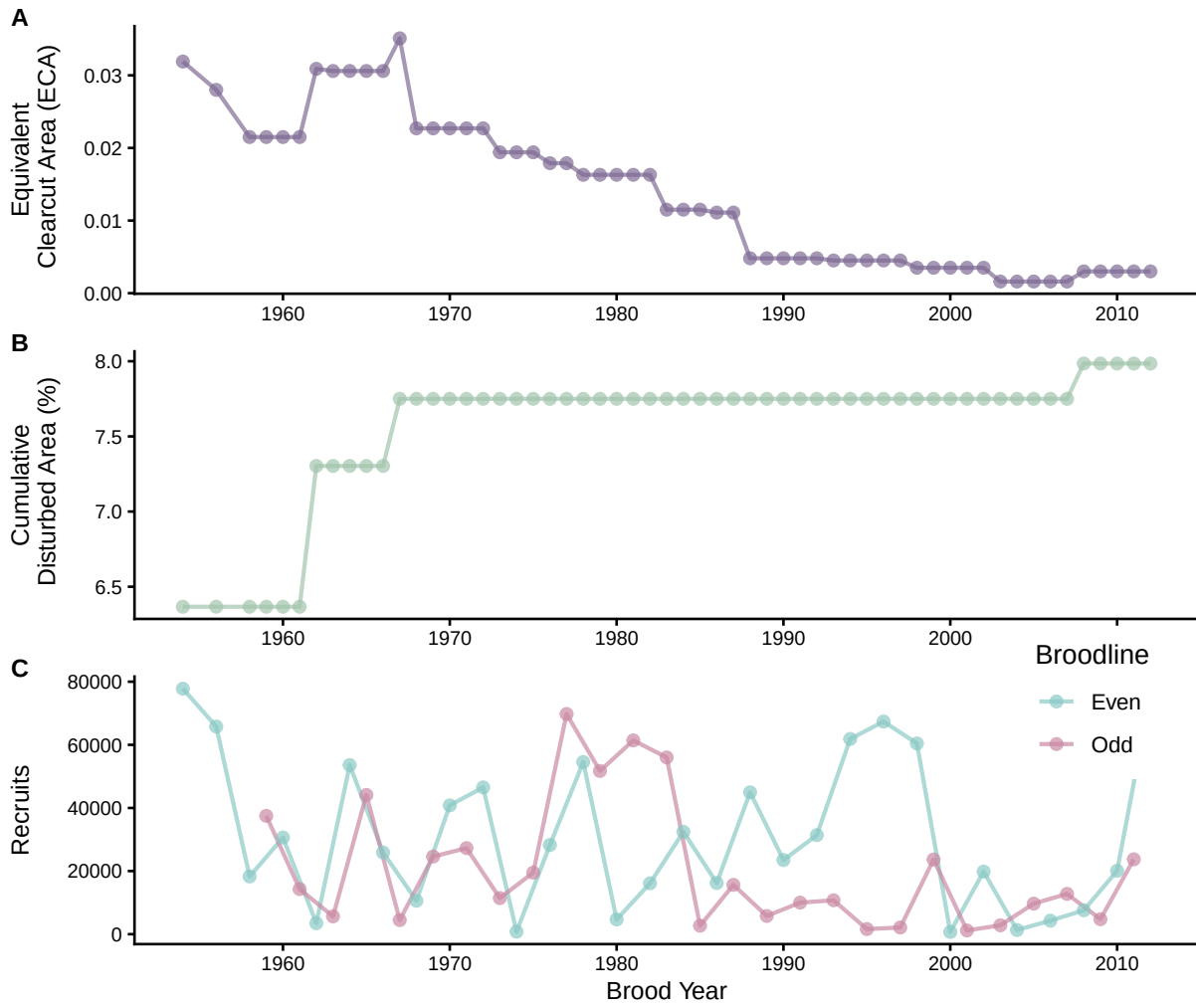
Data for Chum Salmon



Model Results for Chum Salmon

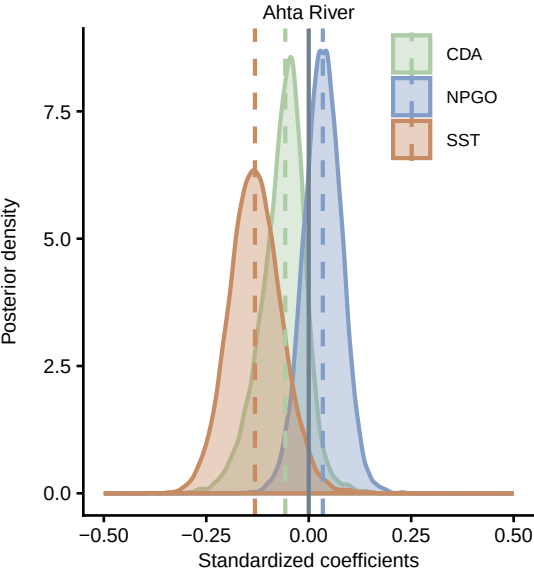


Data for Pink Salmon

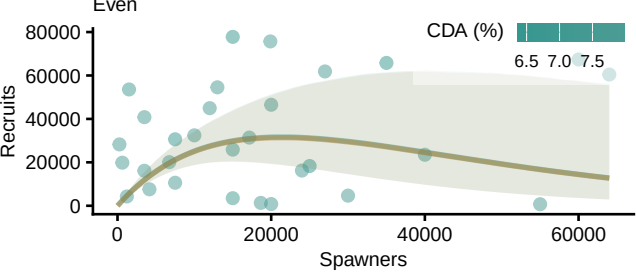


Model Results for Pink Salmon

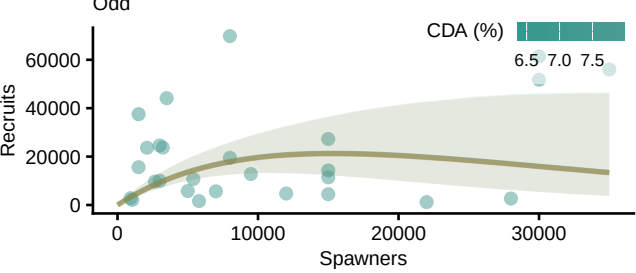
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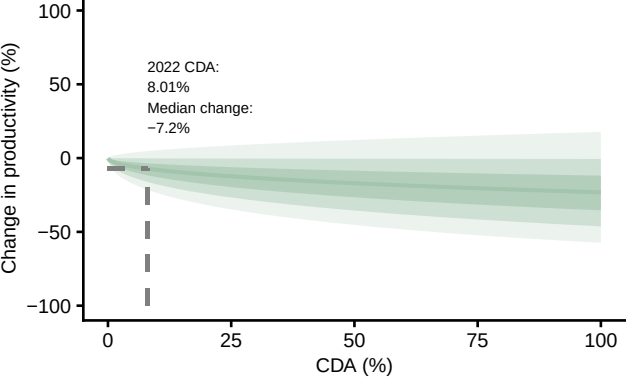
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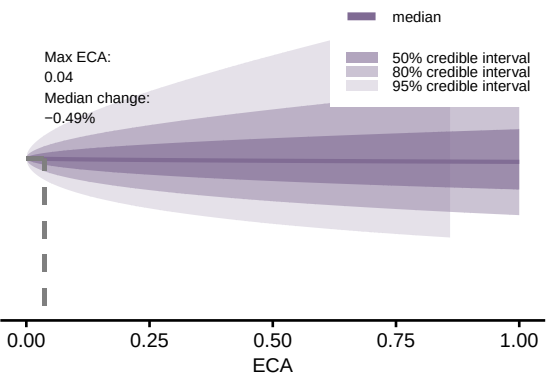
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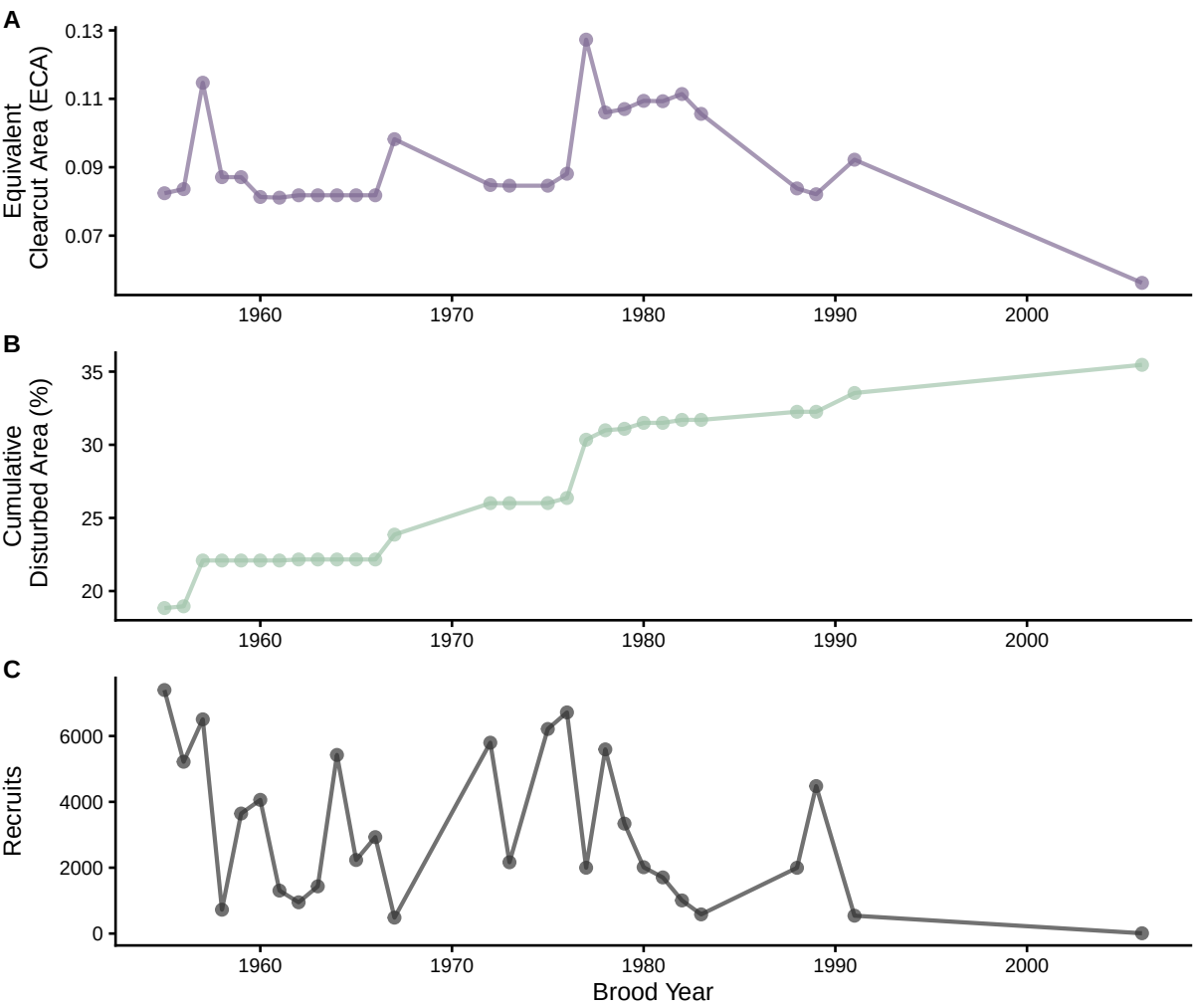


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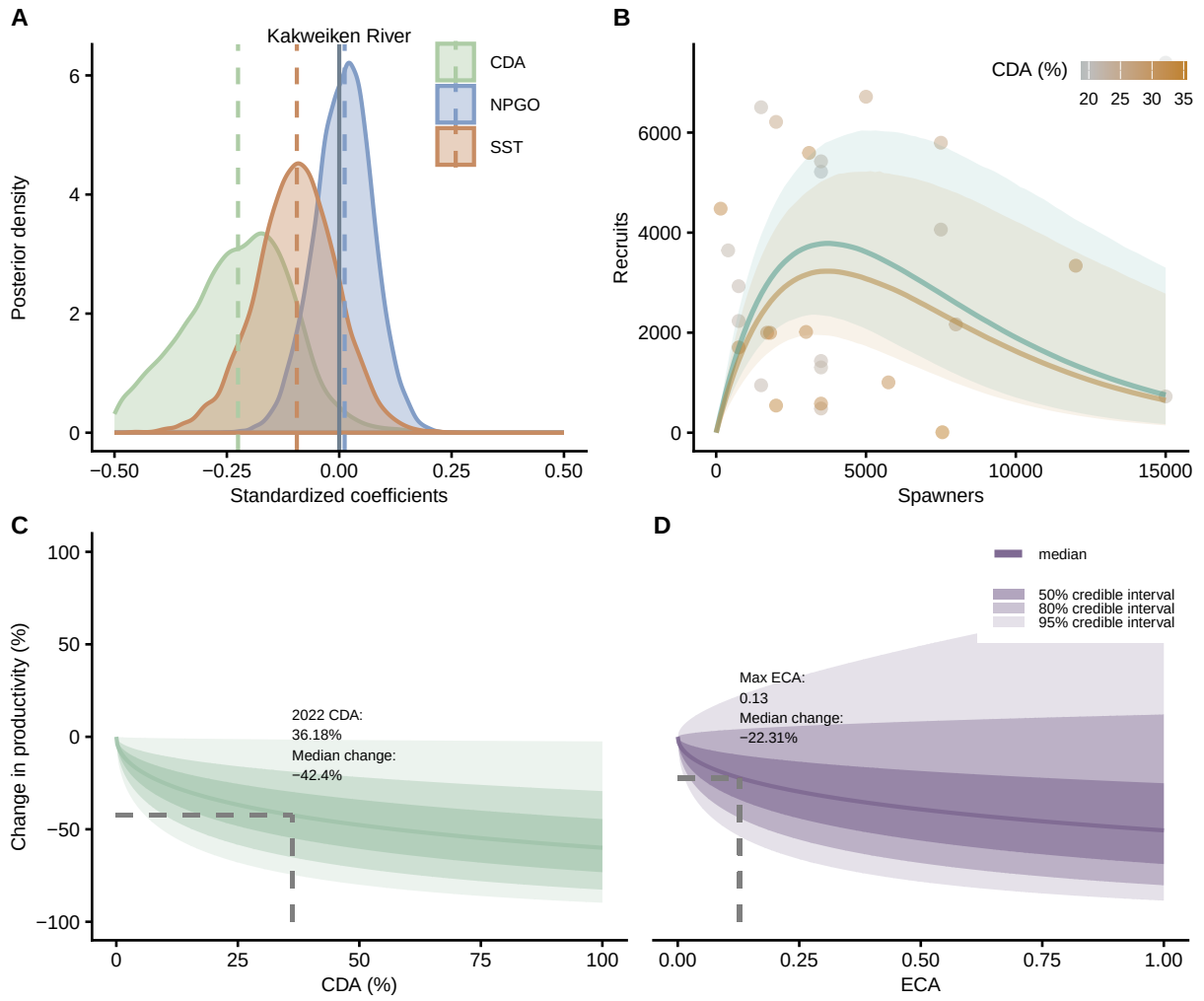


KAKWEIKEN RIVER

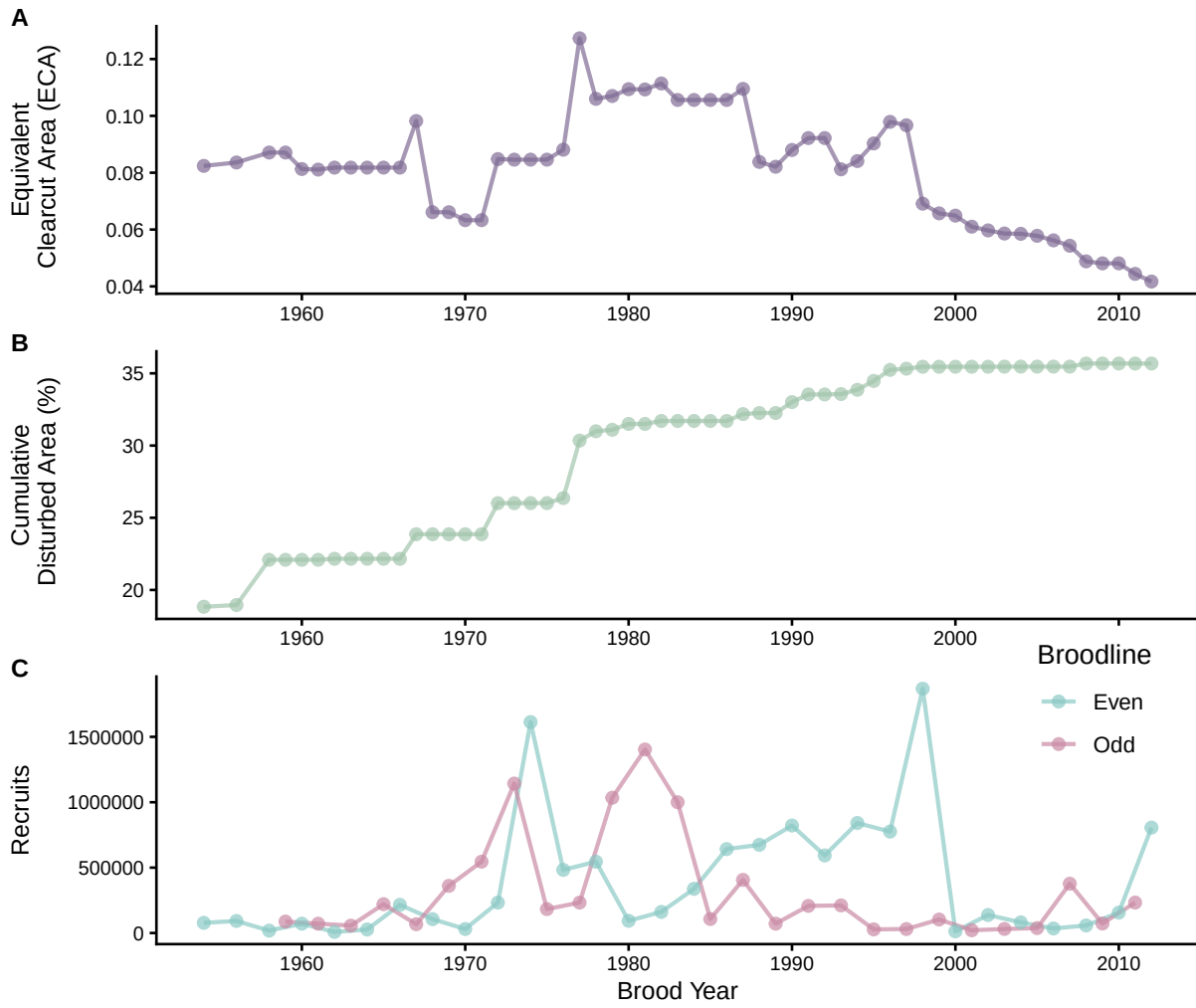
Data for Chum Salmon



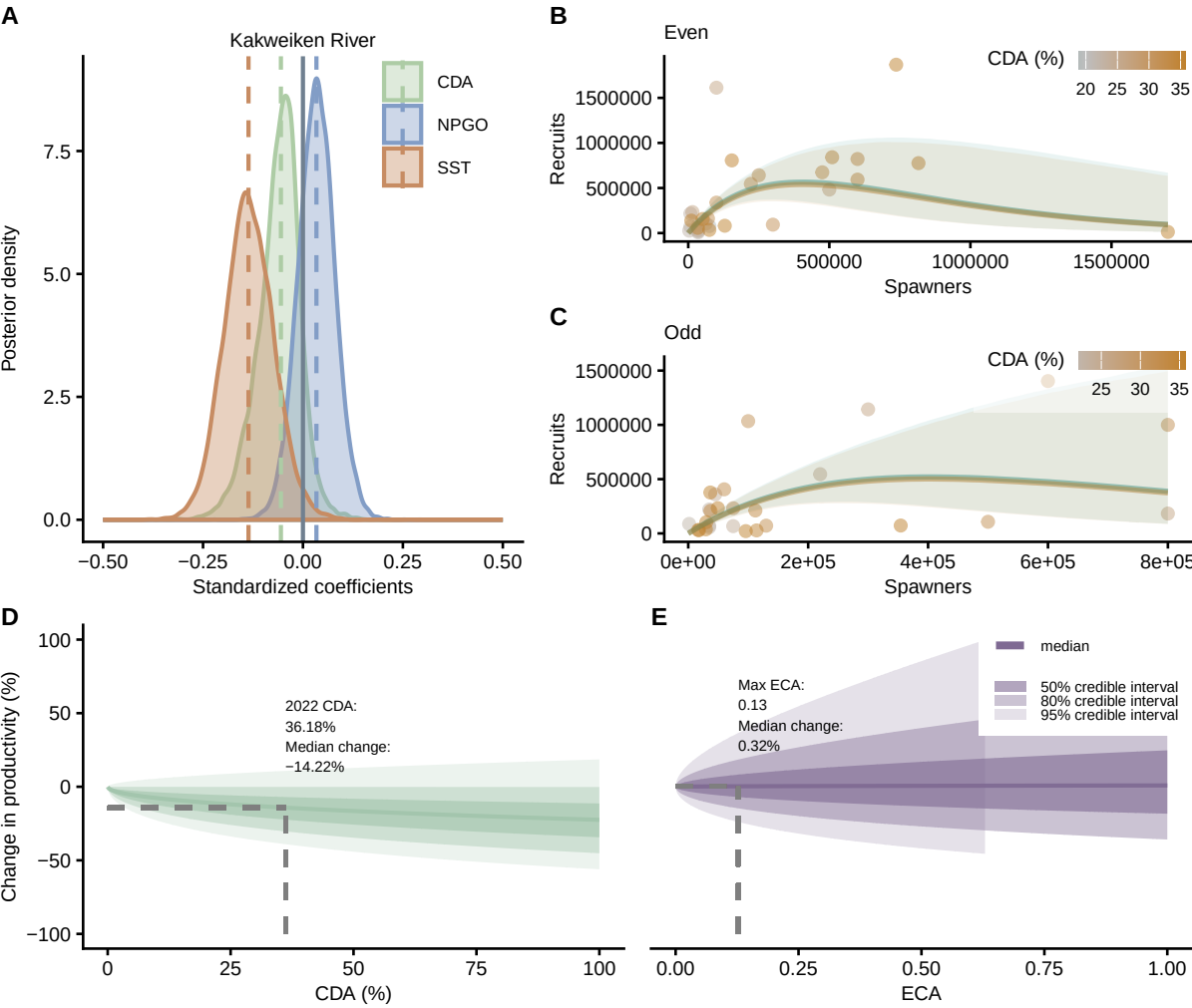
Model Results for Chum Salmon



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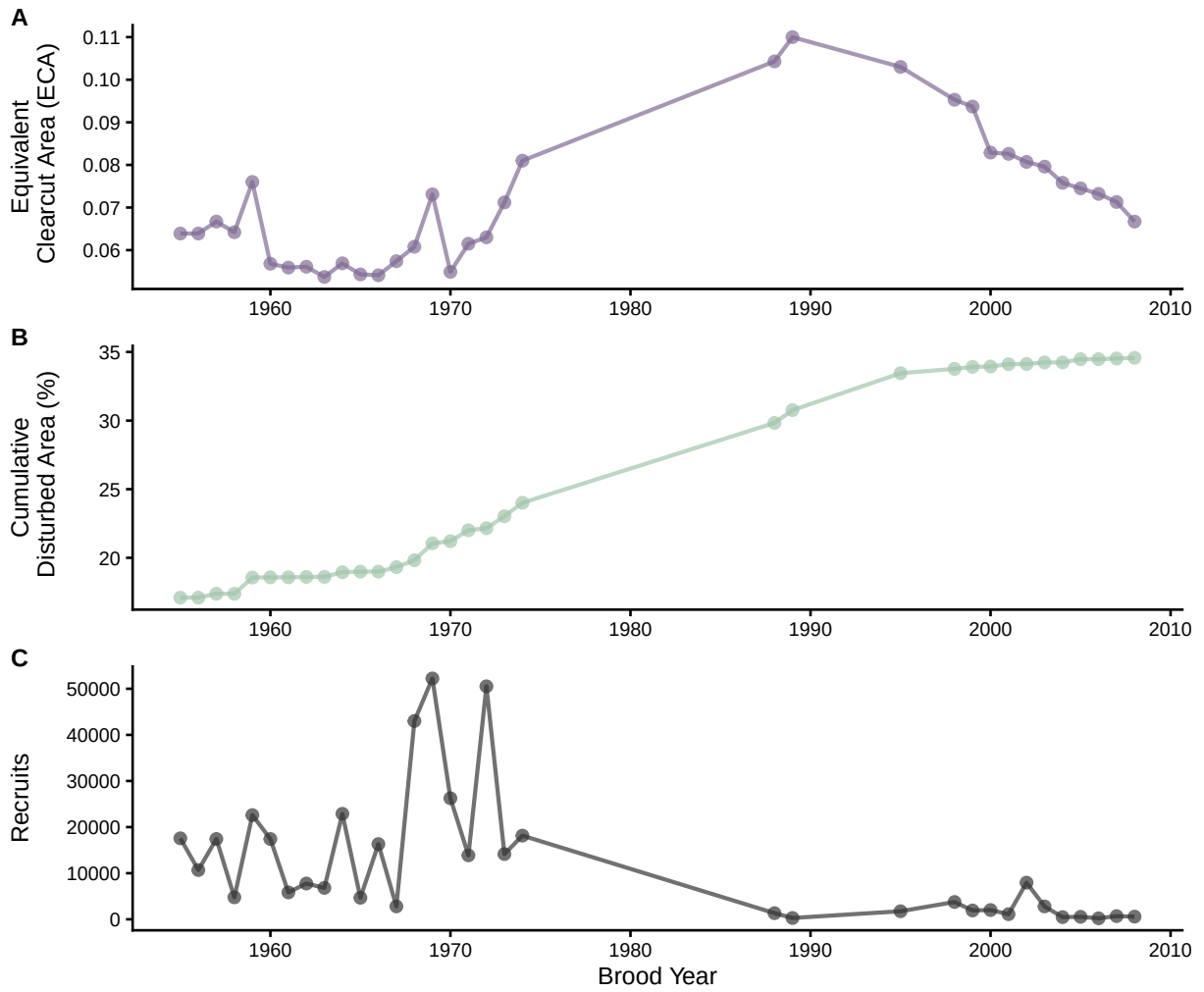


Model Results for Pink Salmon

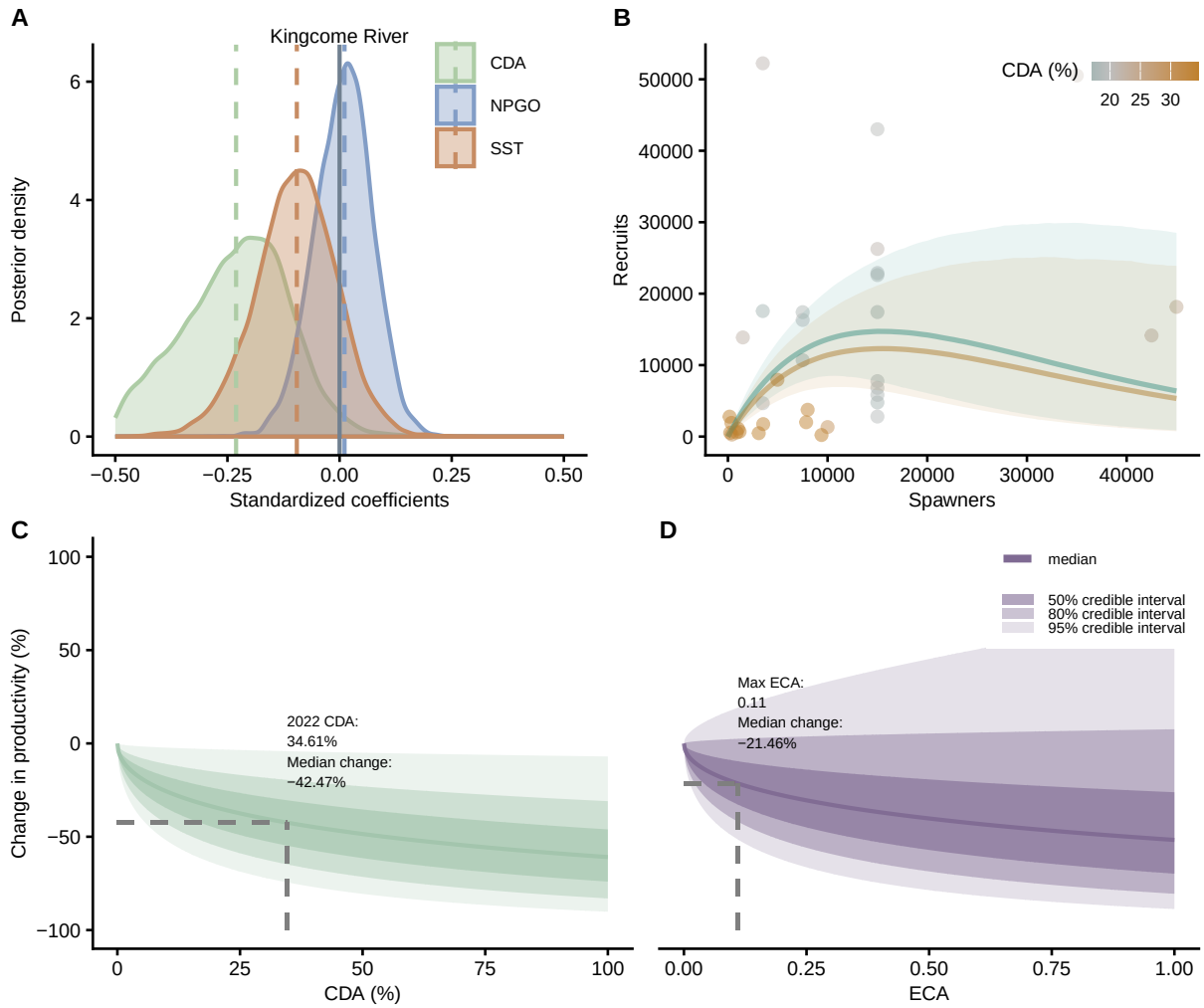


KINGCOME RIVER

Data for Chum Salmon

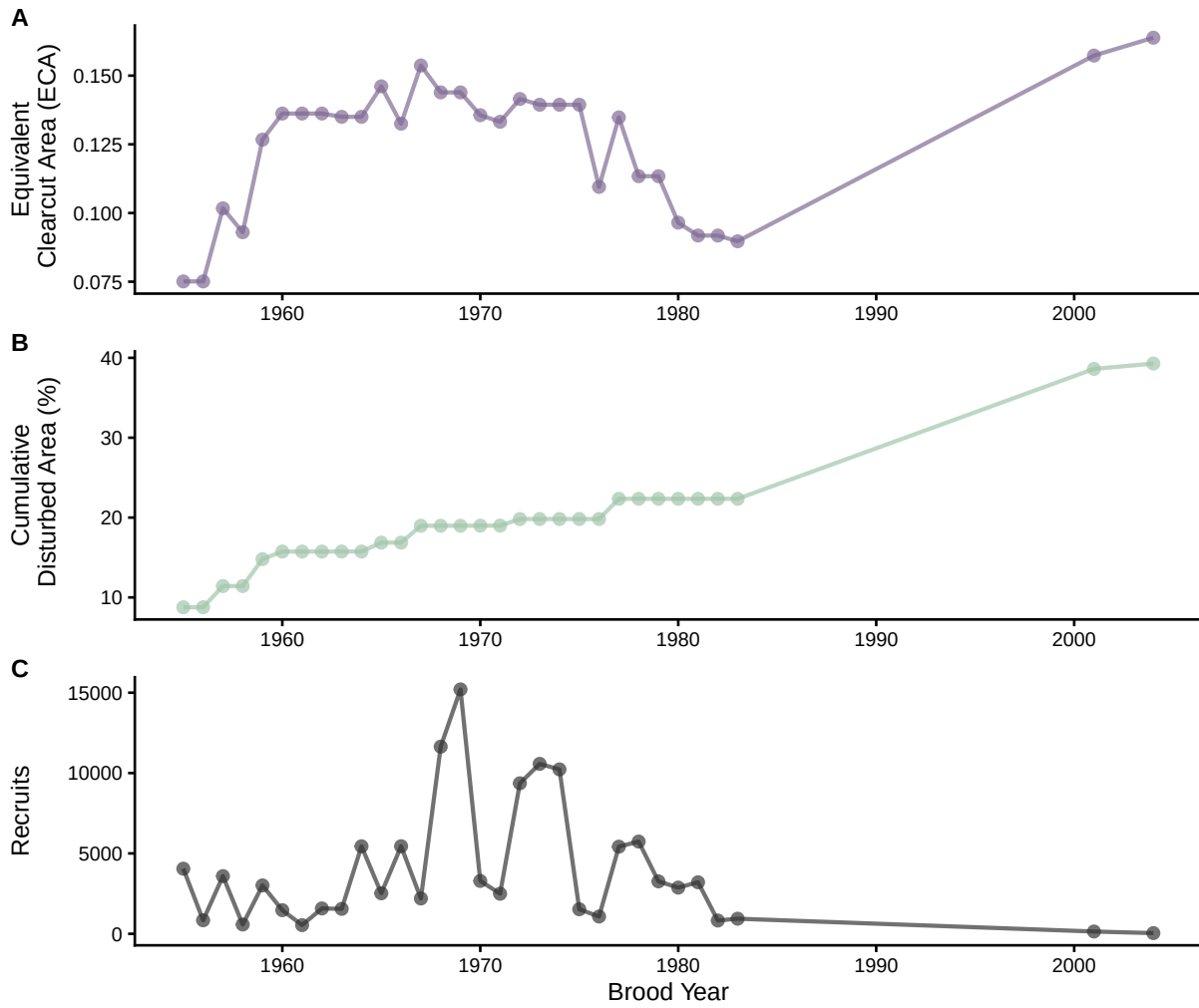


Model Results for Chum Salmon

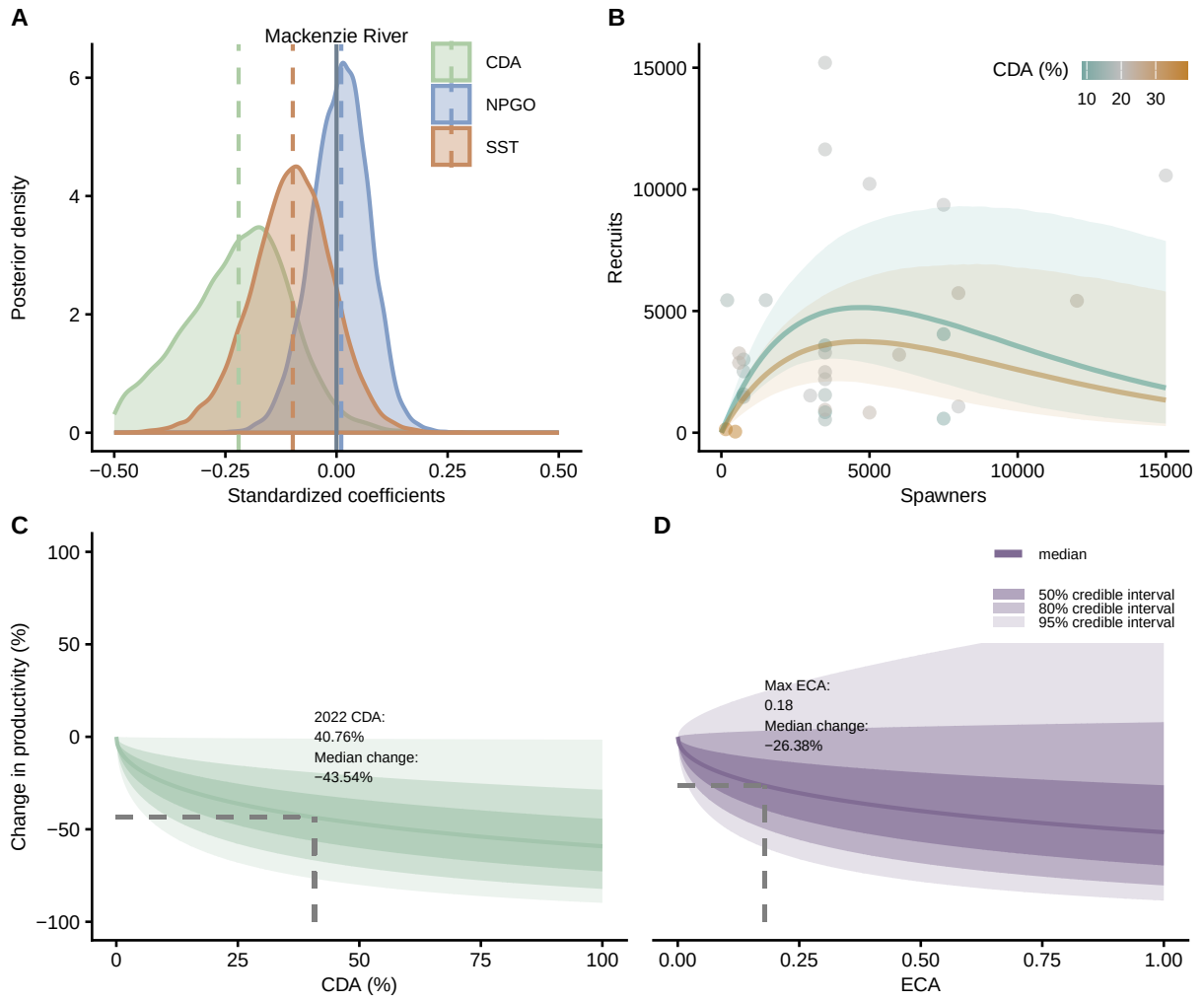


MACKENZIE RIVER

Data for Chum Salmon

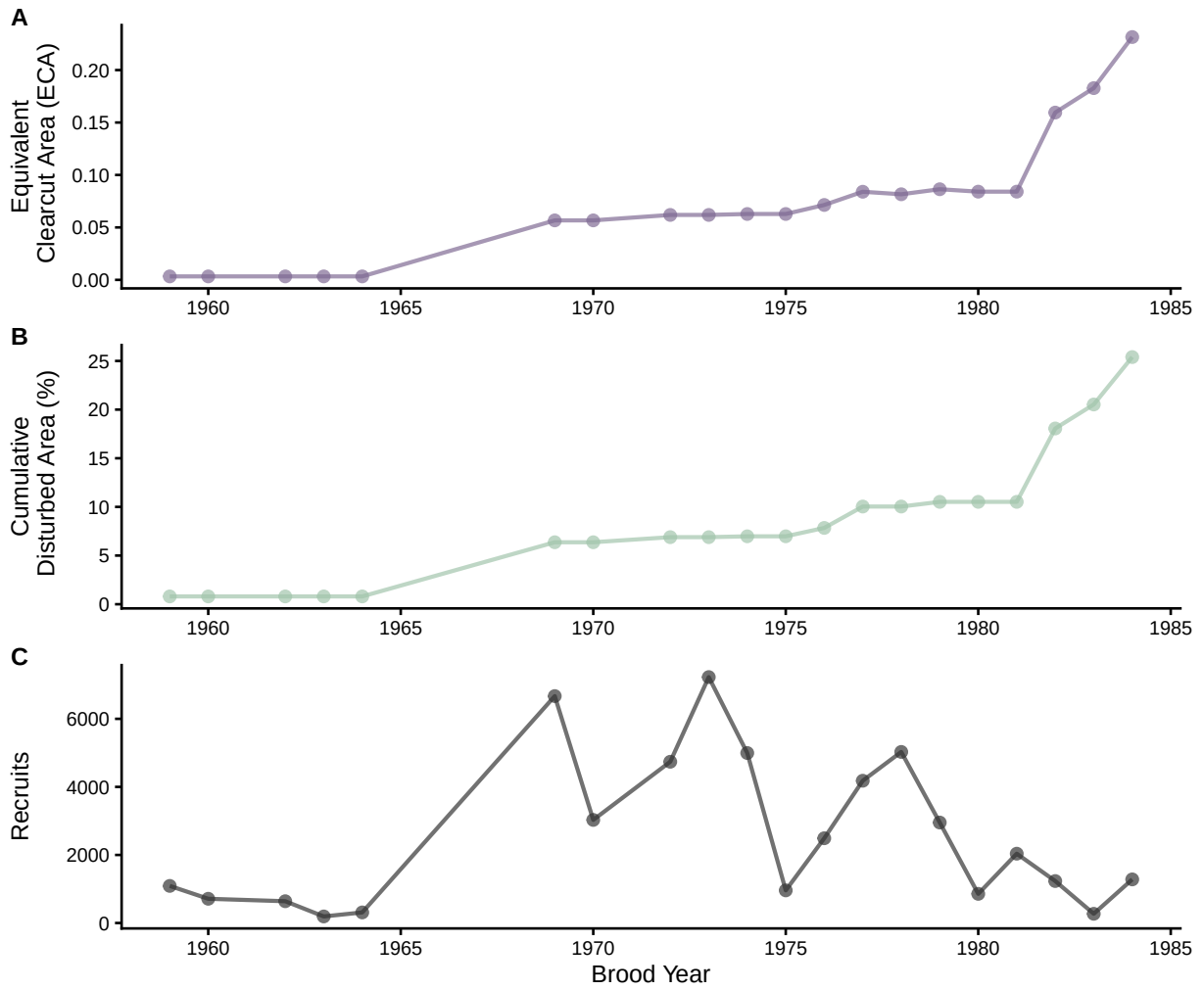


Model Results for Chum Salmon

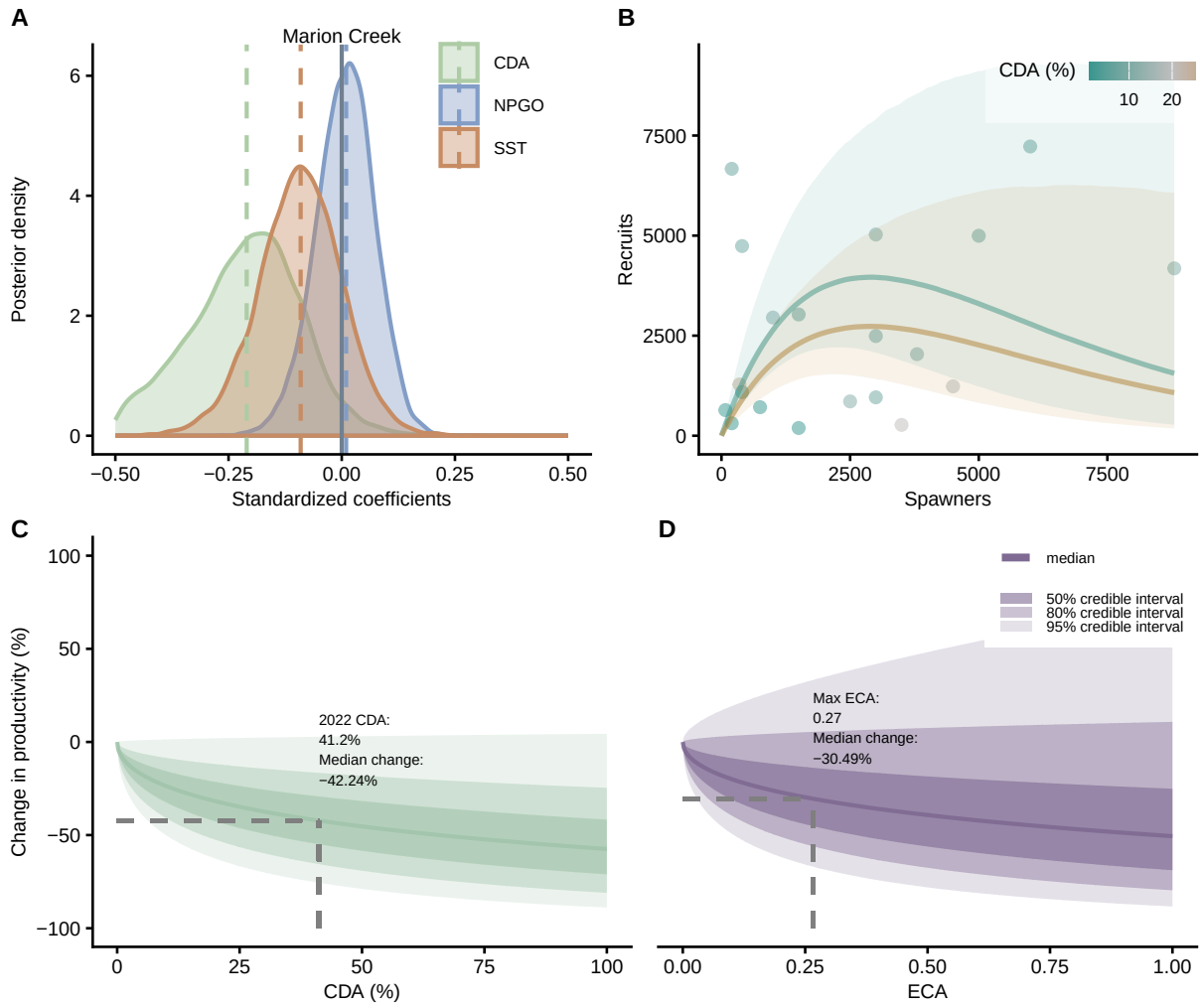


MARION CREEK

Data for Chum Salmon

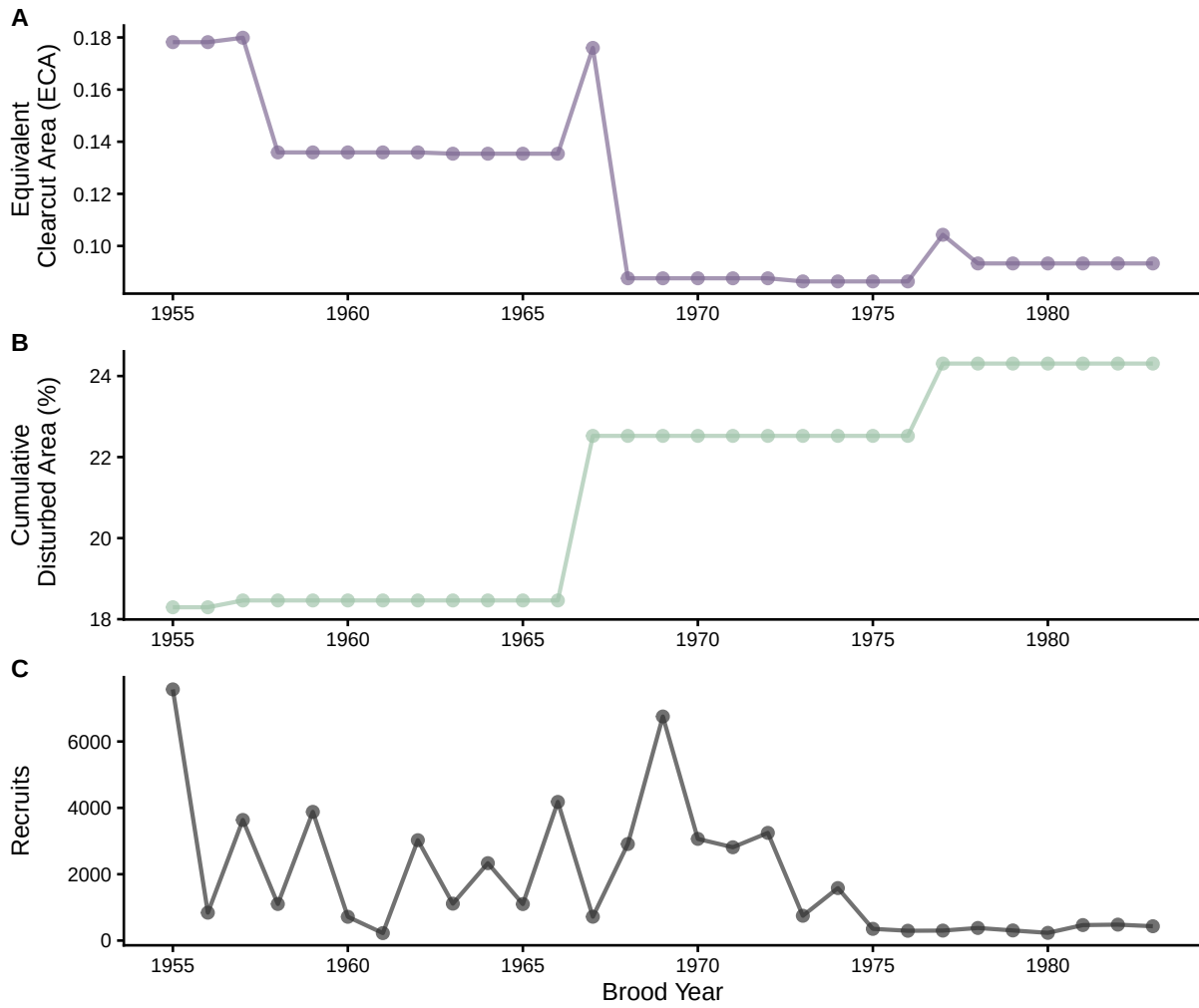


Model Results for Chum Salmon

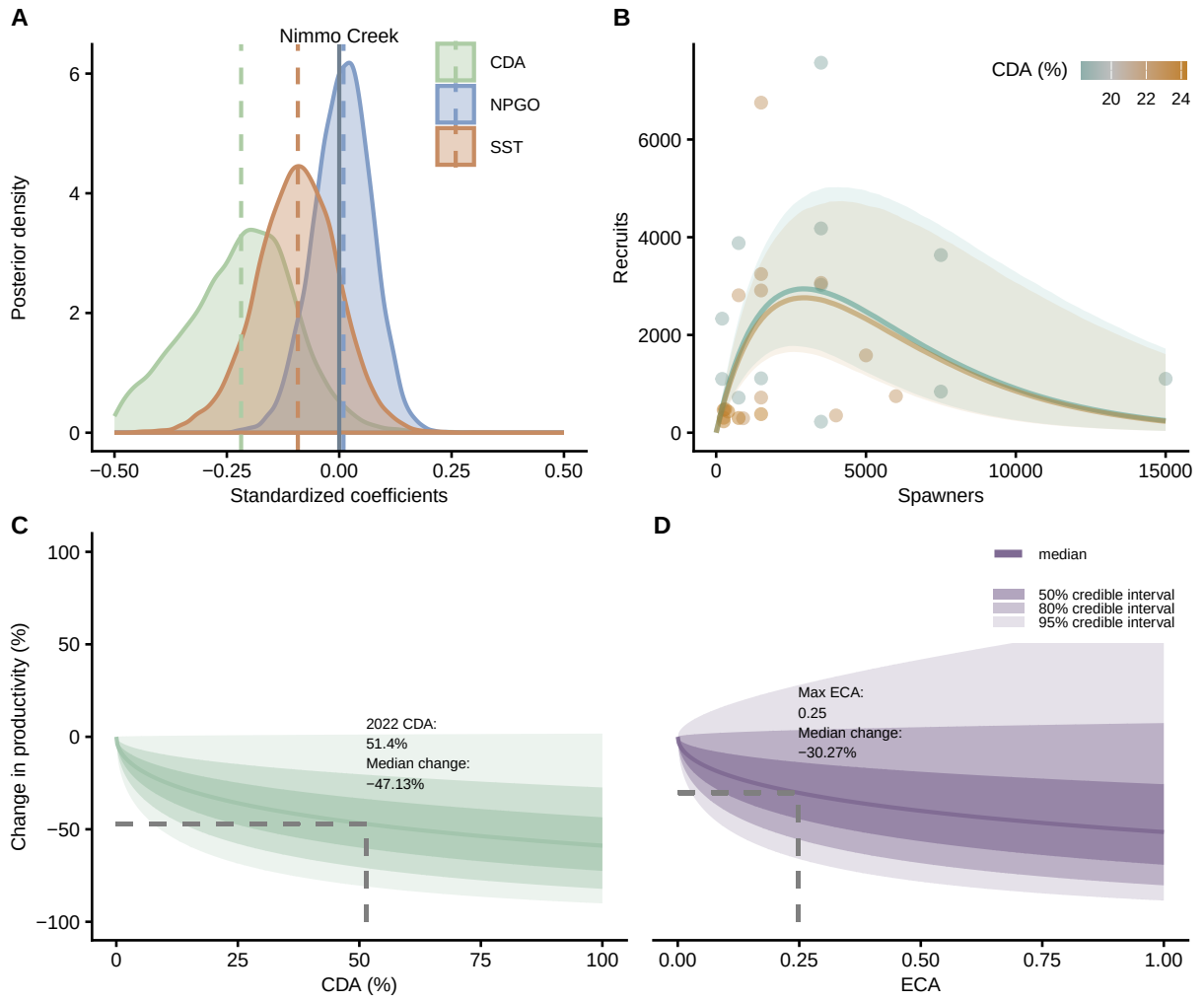


NIMMO CREEK

Data for Chum Salmon

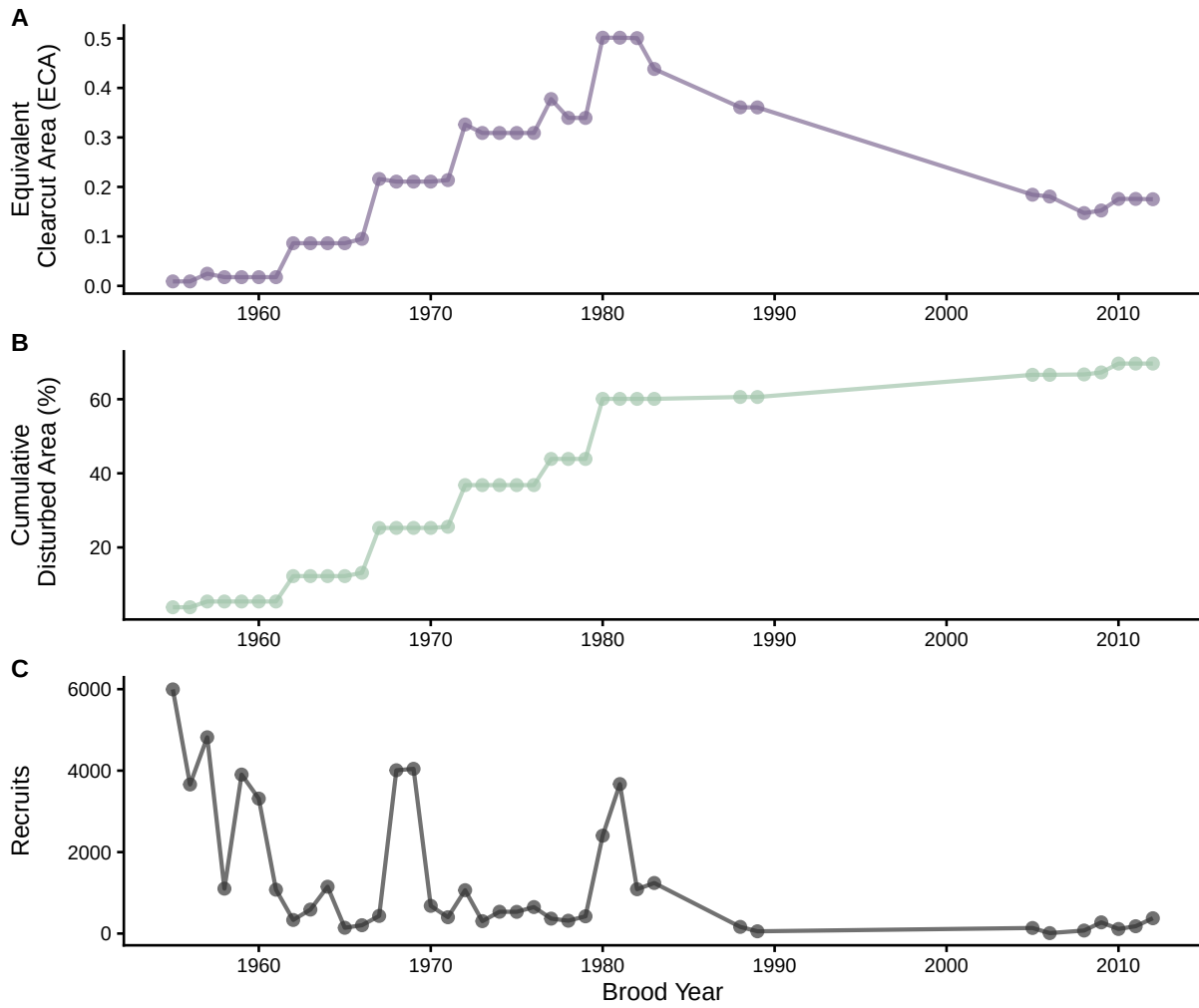


Model Results for Chum Salmon

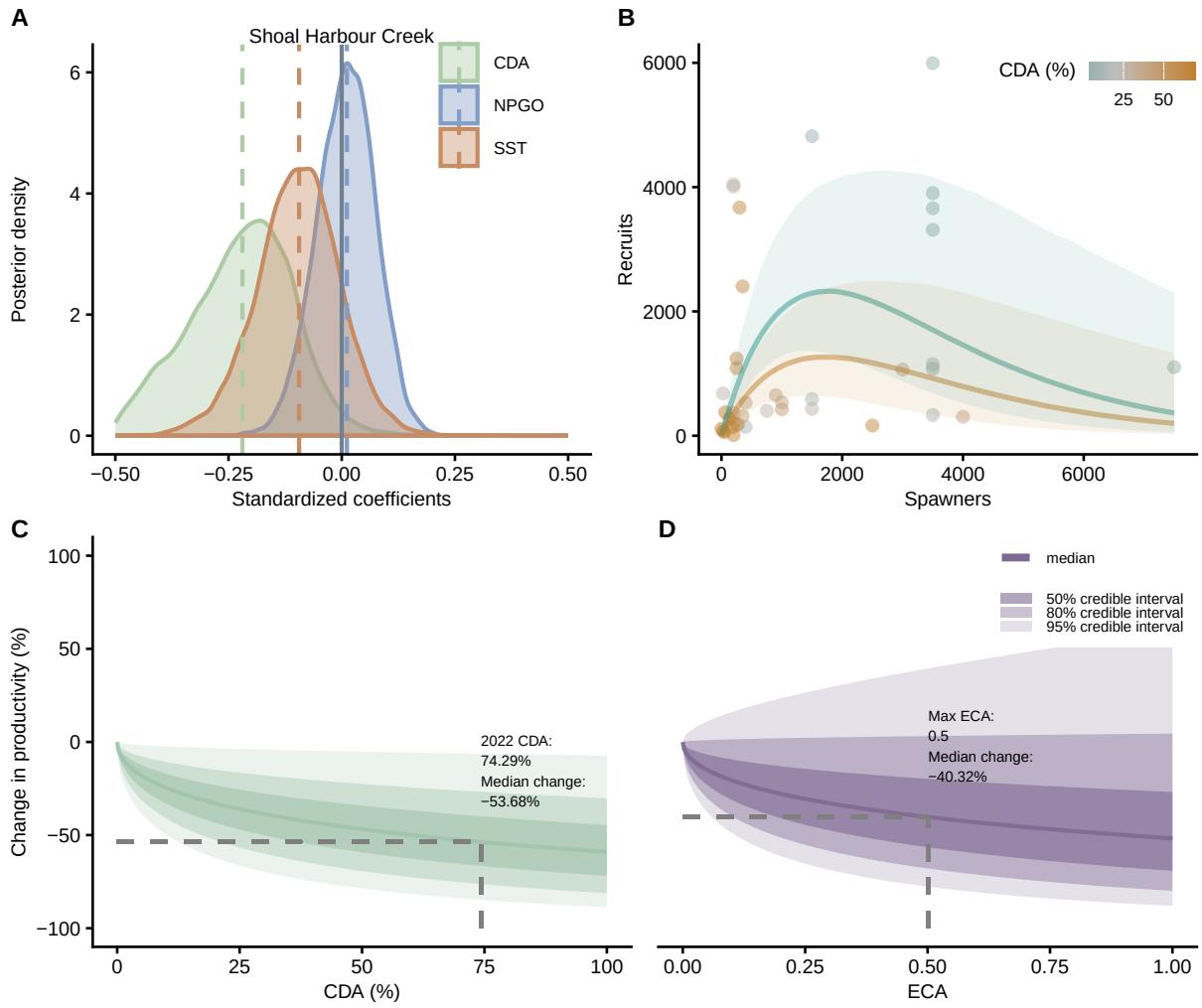


SHOAL HARBOUR CREEK

Data for Chum Salmon

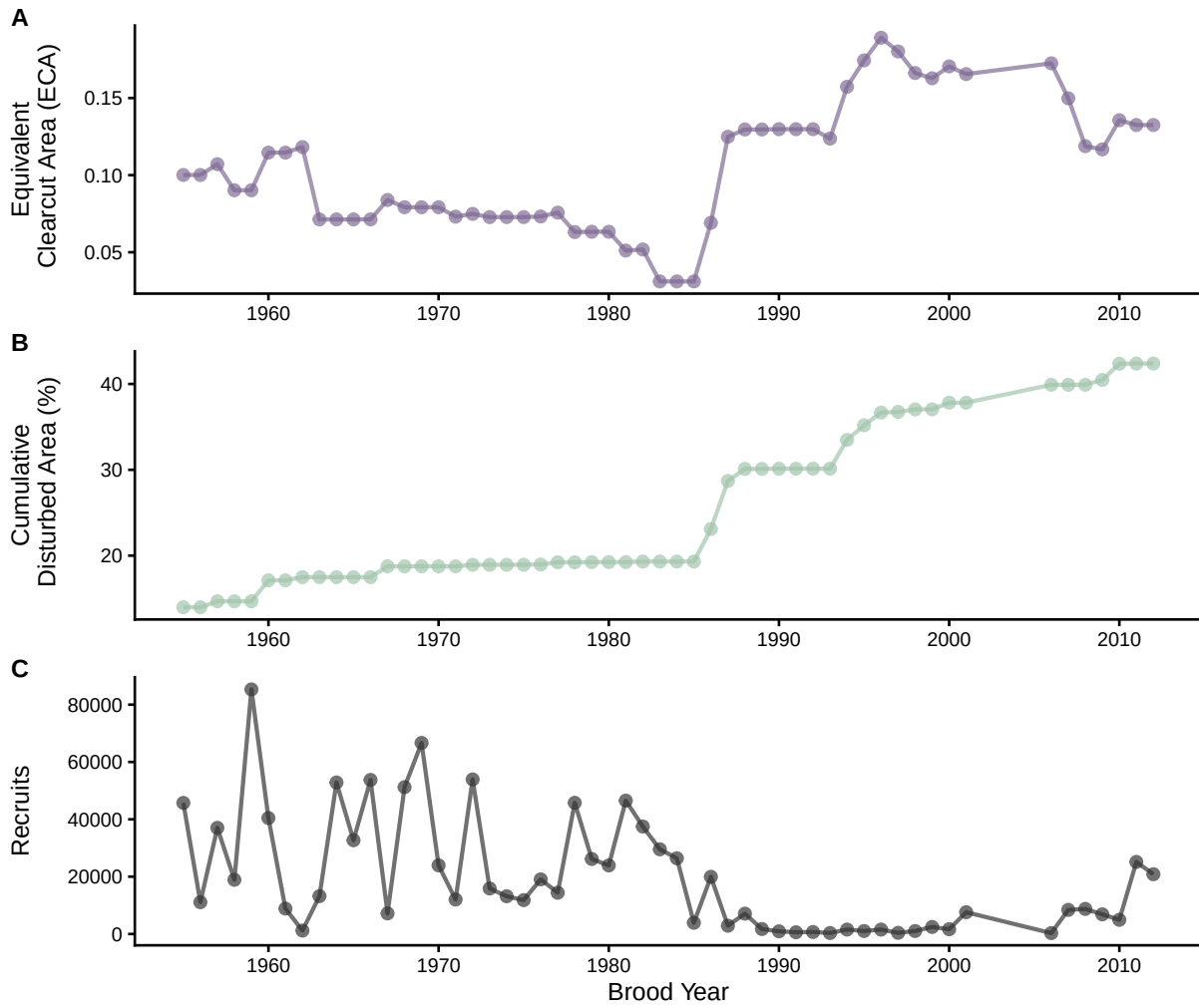


Model Results for Chum Salmon

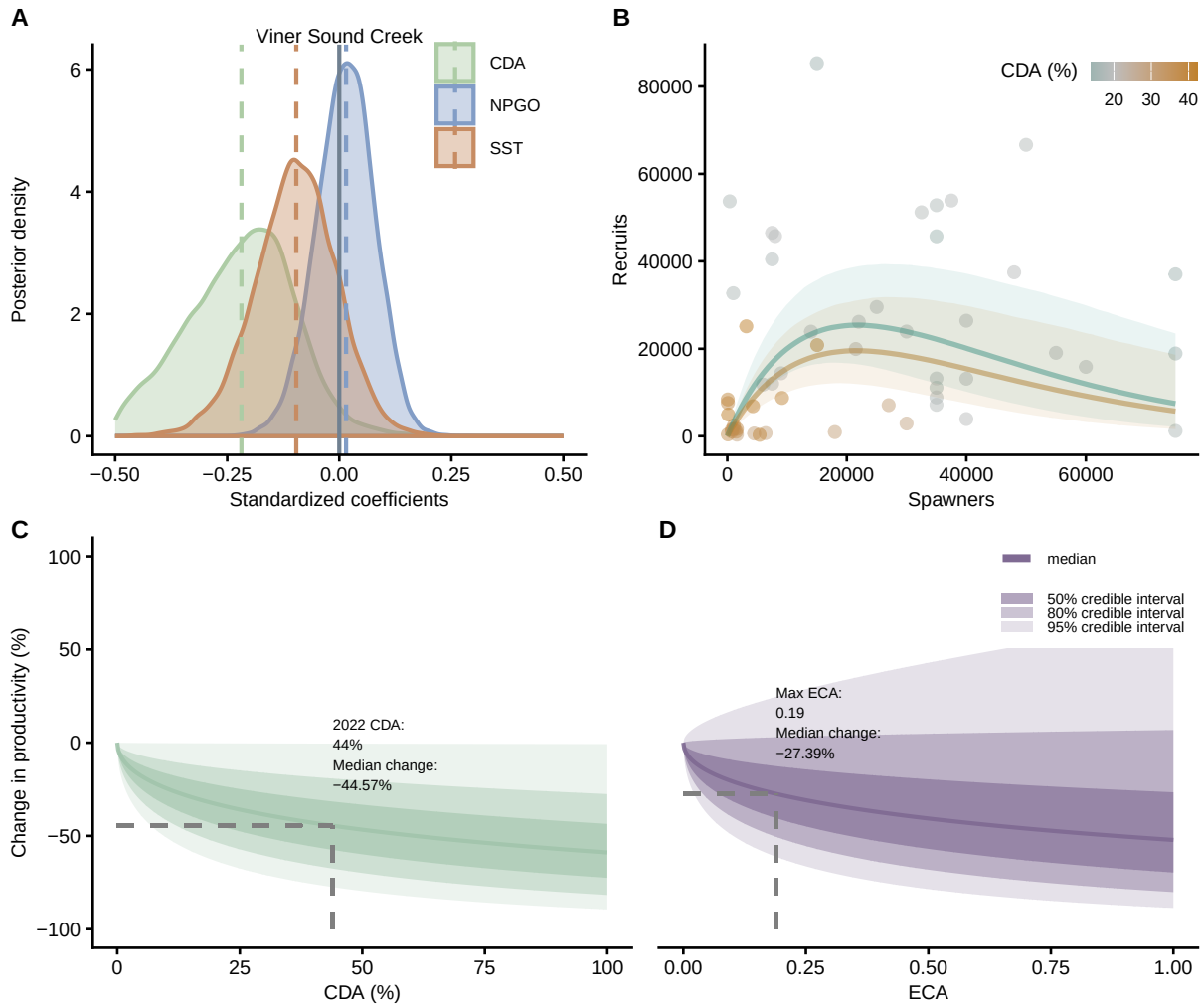


VINER SOUND CREEK

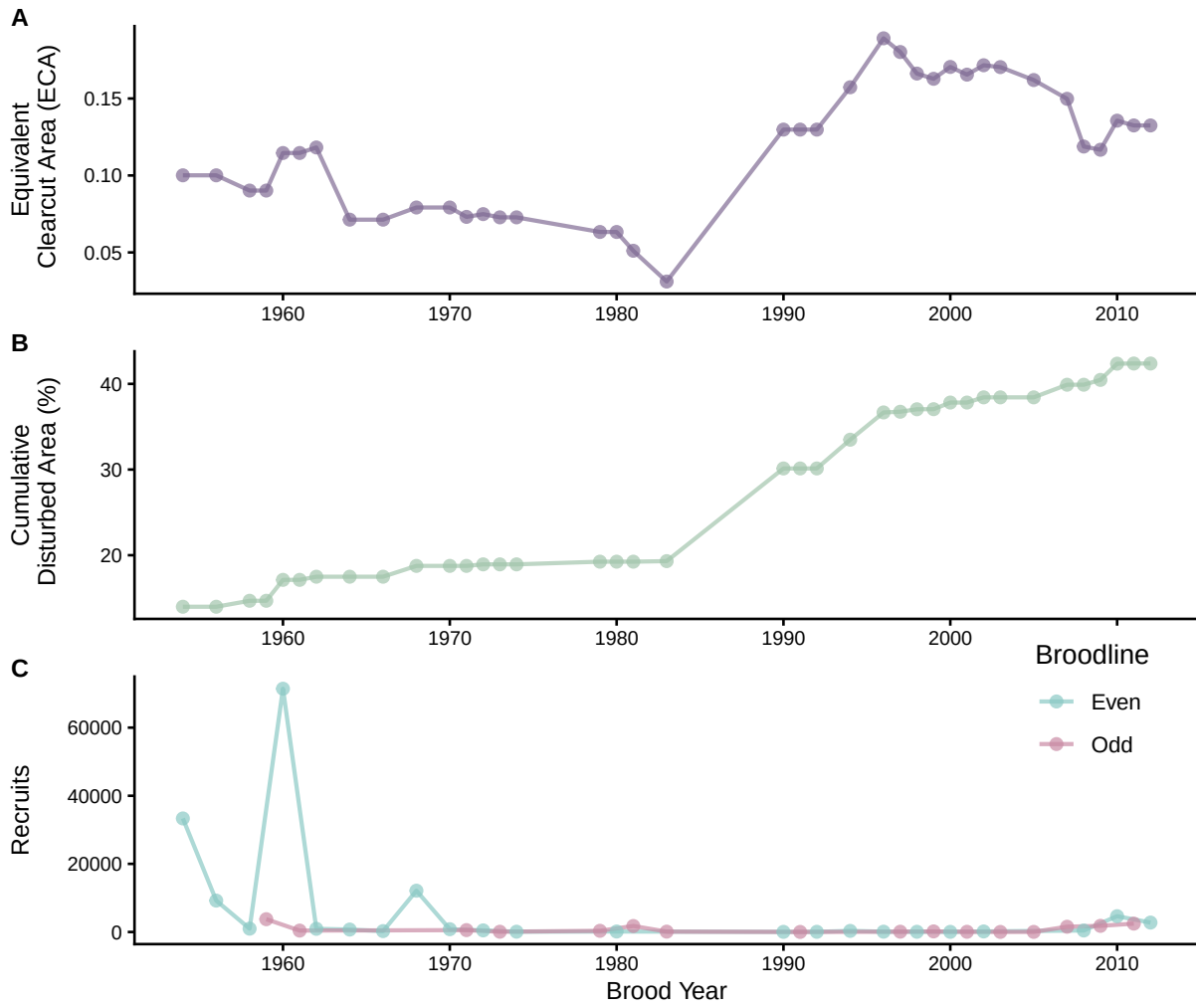
Data for Chum Salmon



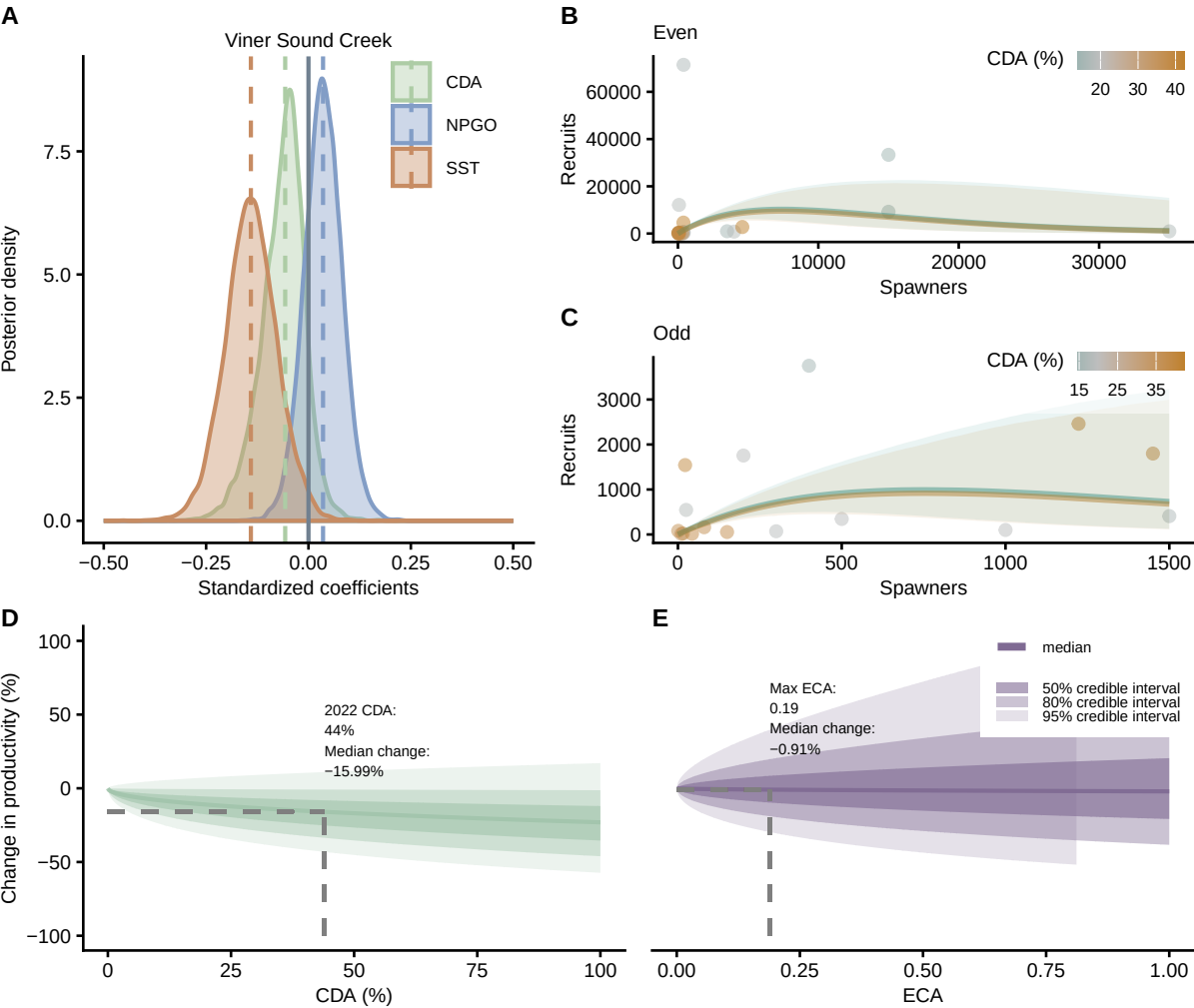
Model Results for Chum Salmon



Data for Pink Salmon

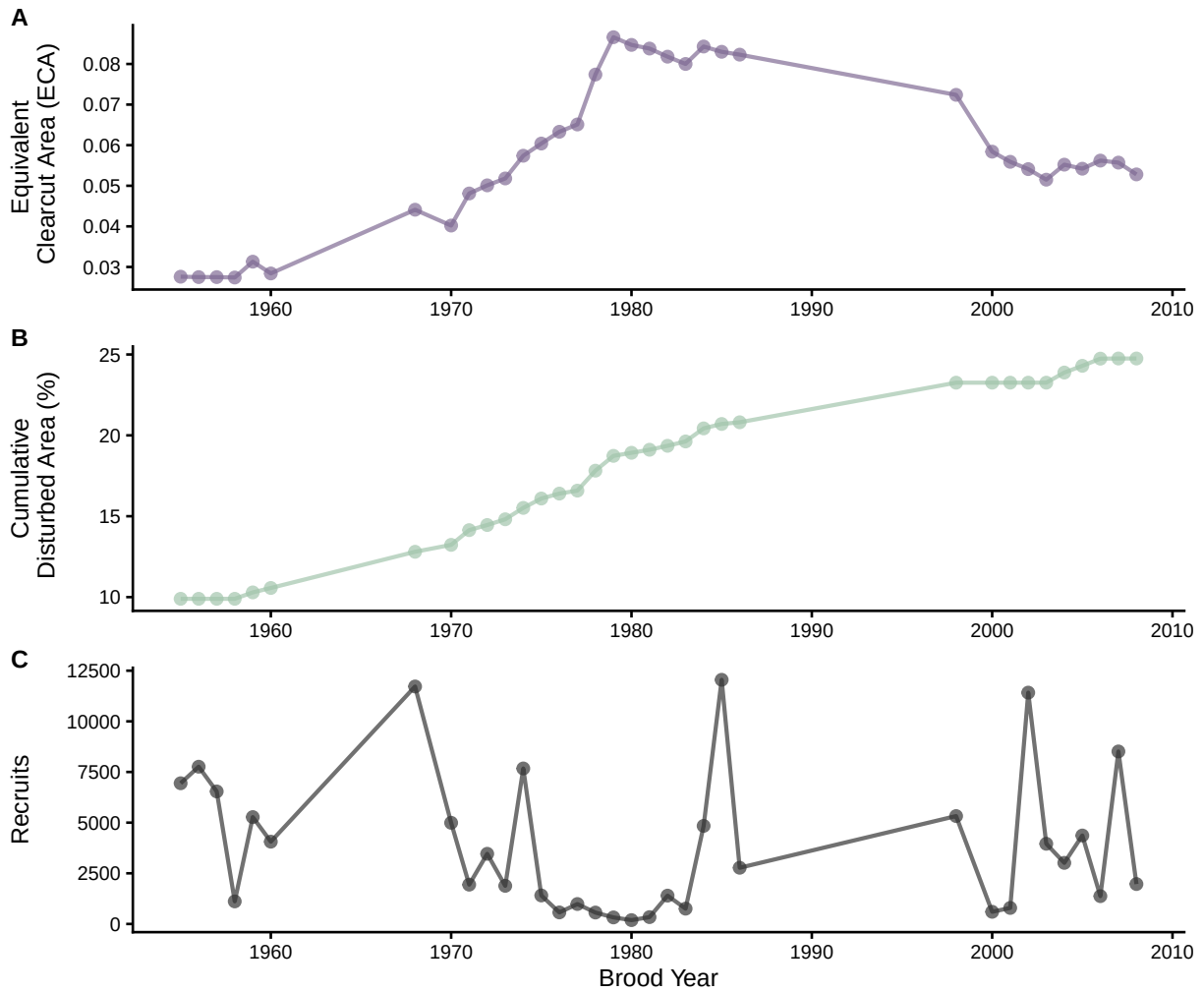


Model Results for Pink Salmon

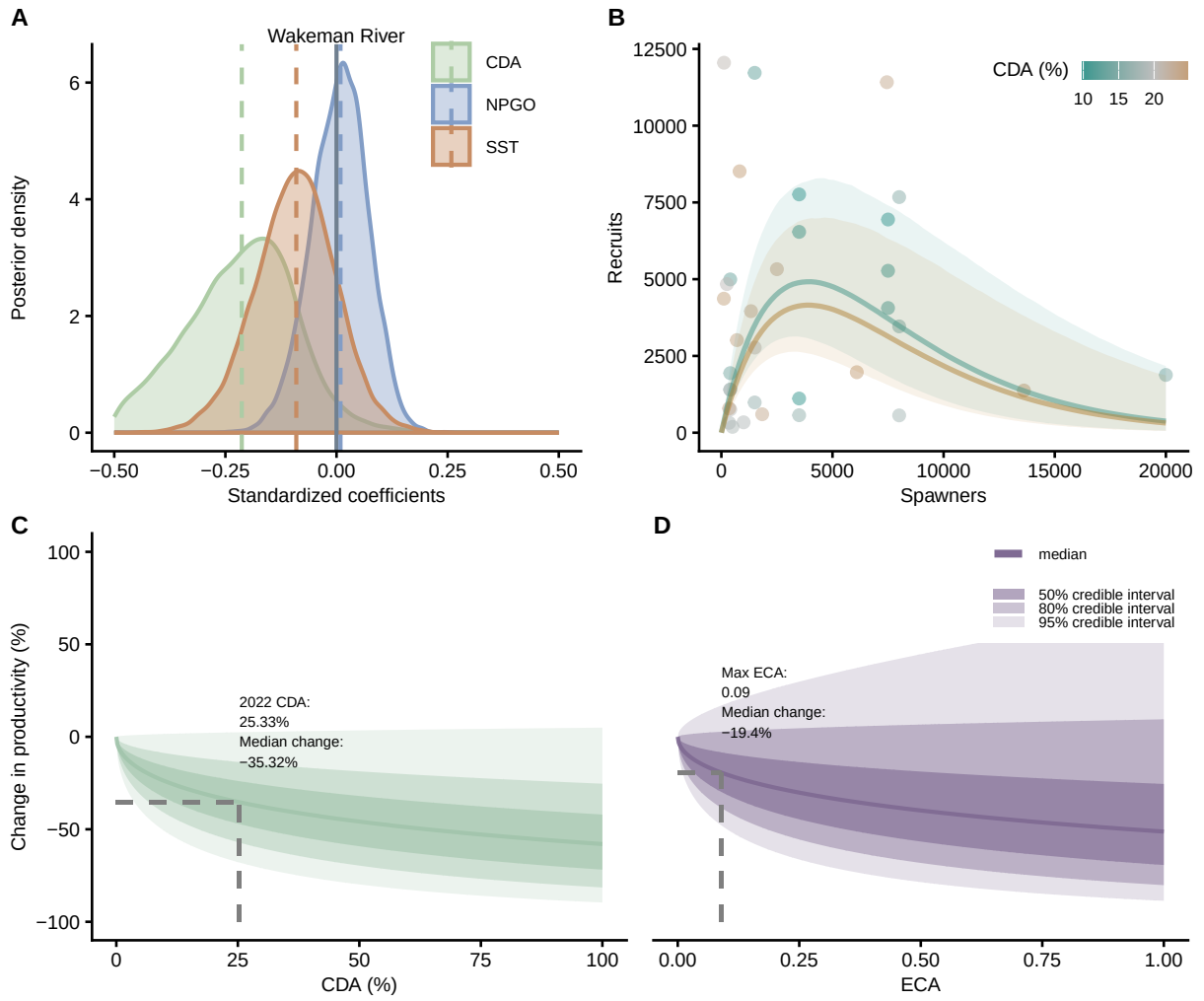


WAKEMAN RIVER

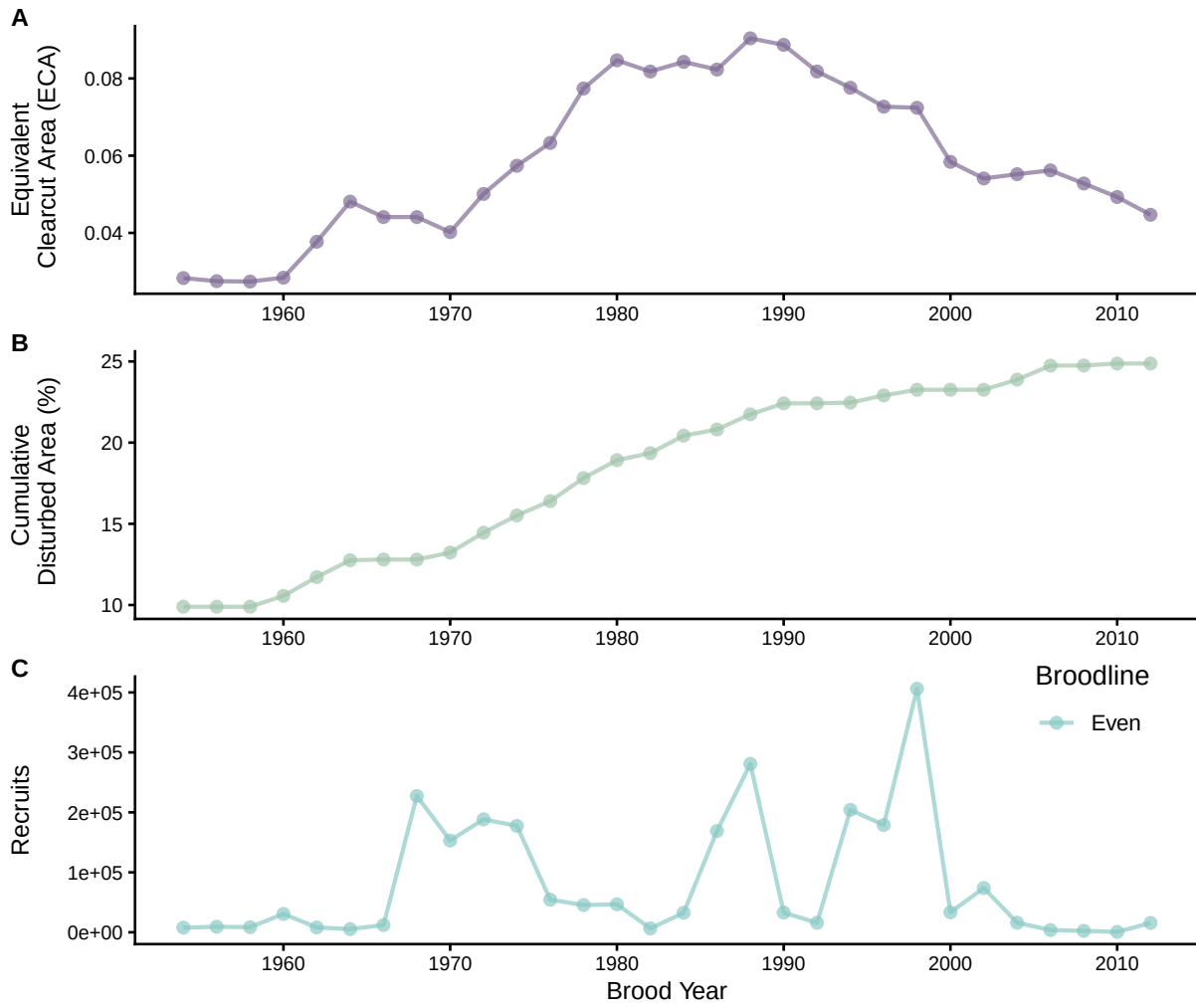
Data for Chum Salmon



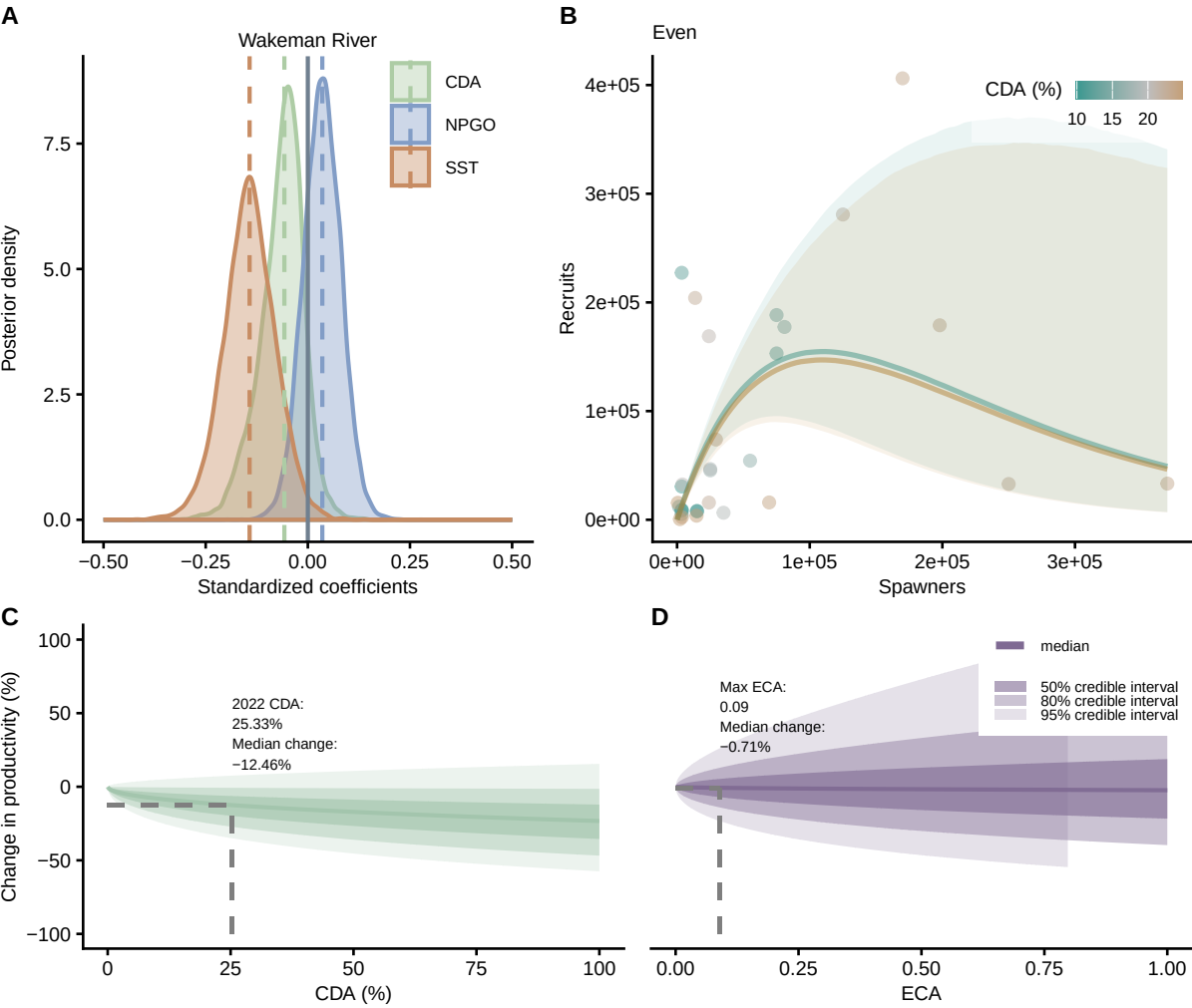
Model Results for Chum Salmon



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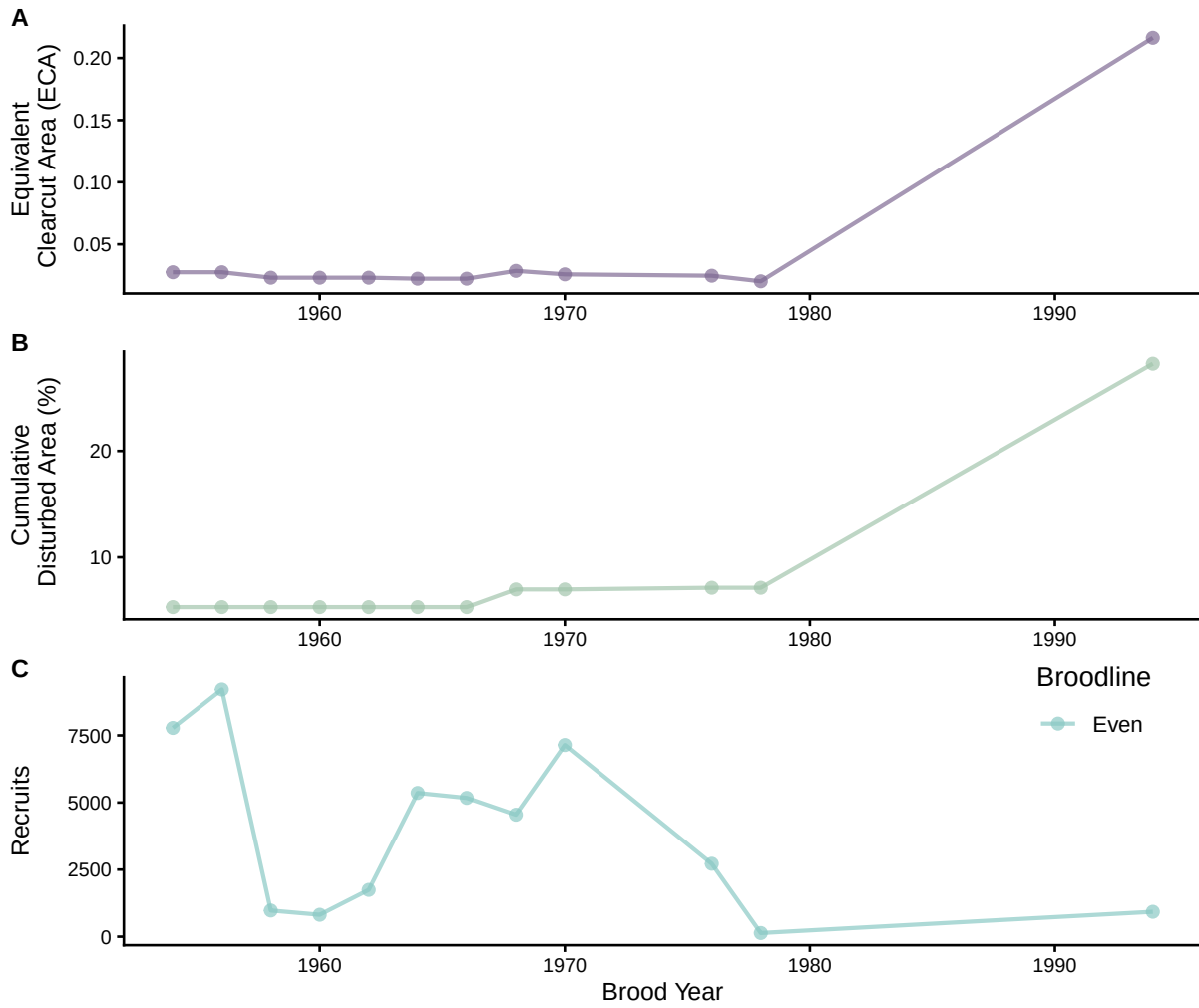


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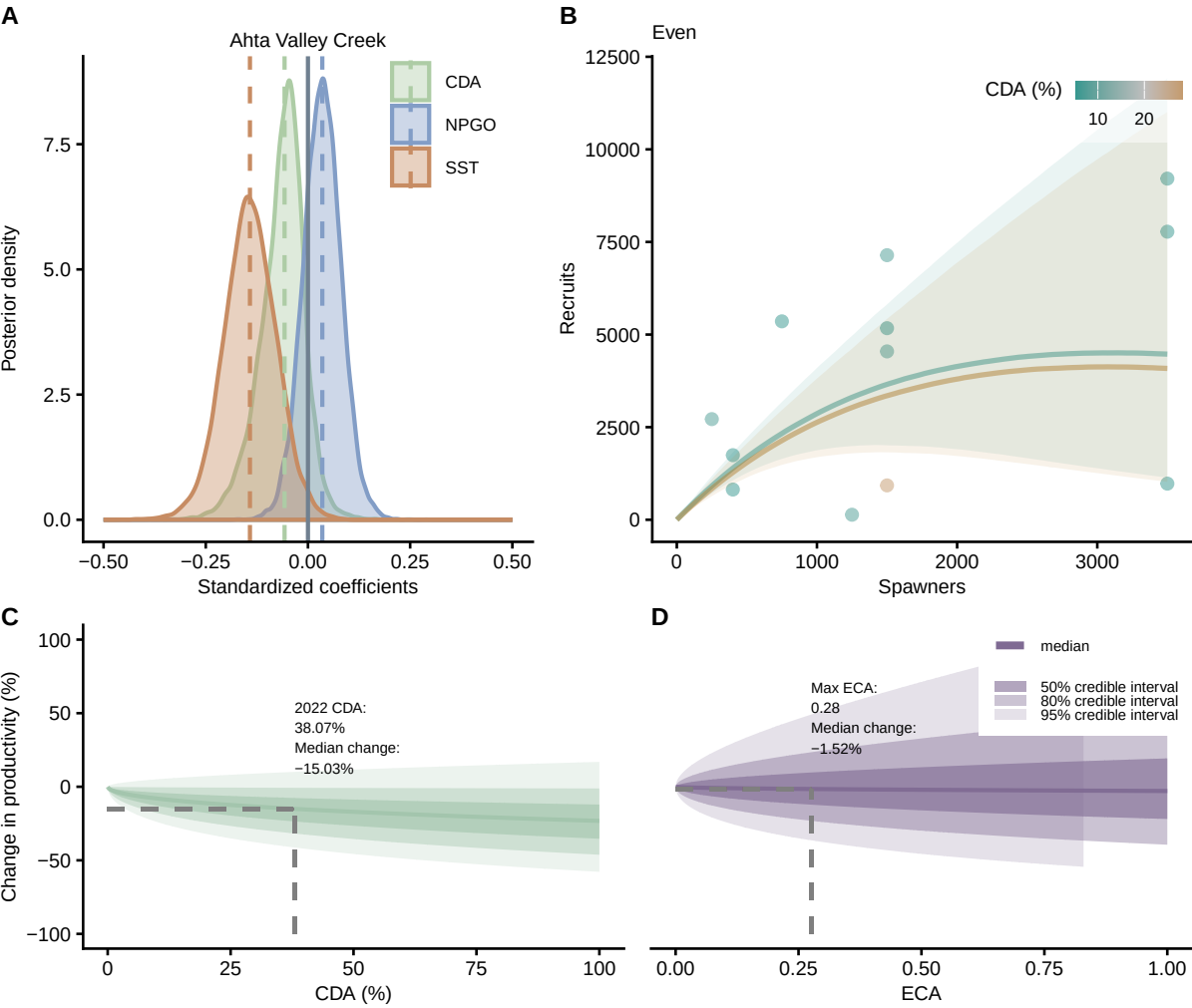


AHTA VALLEY CREEK

Data for Pink Salmon

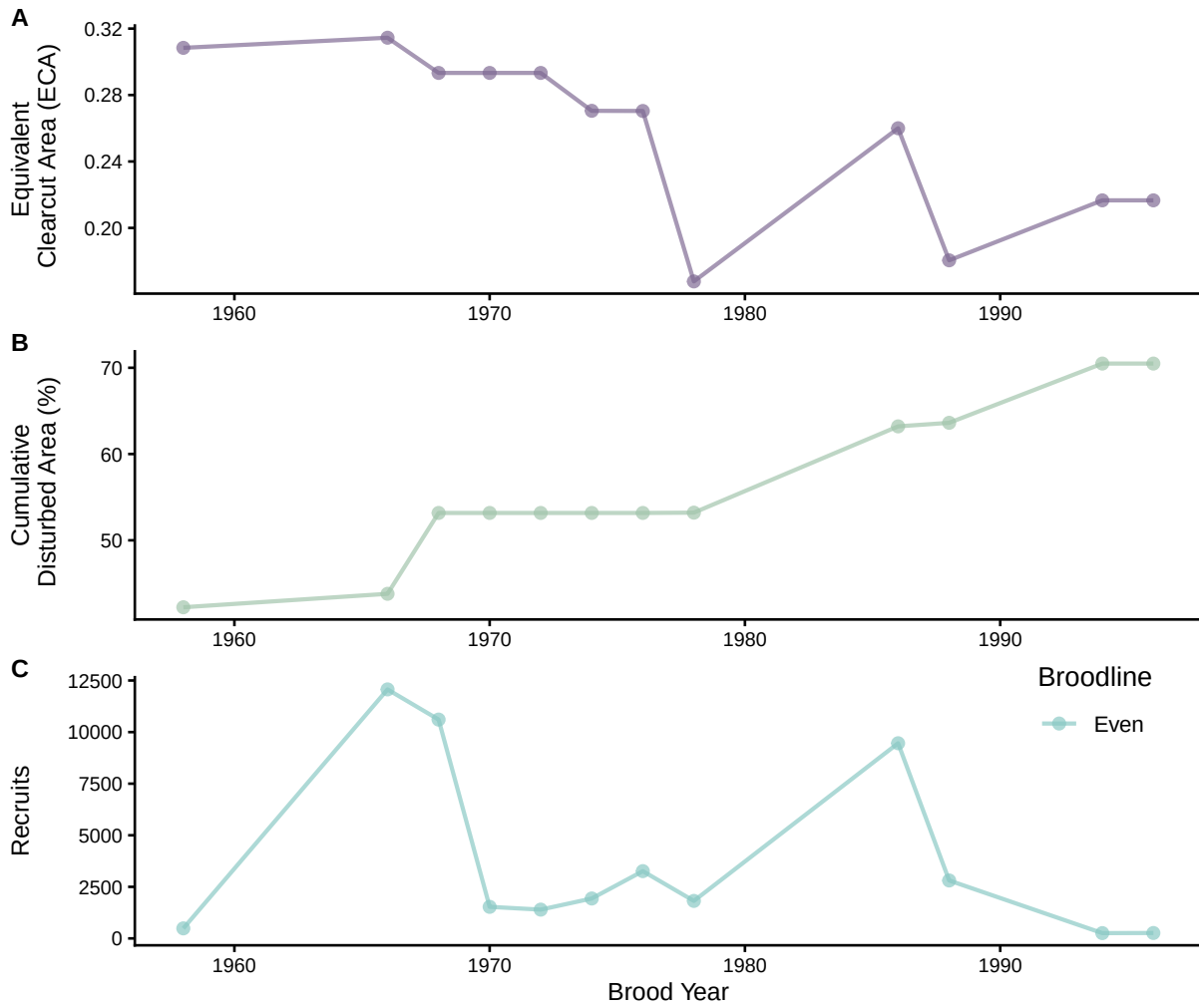


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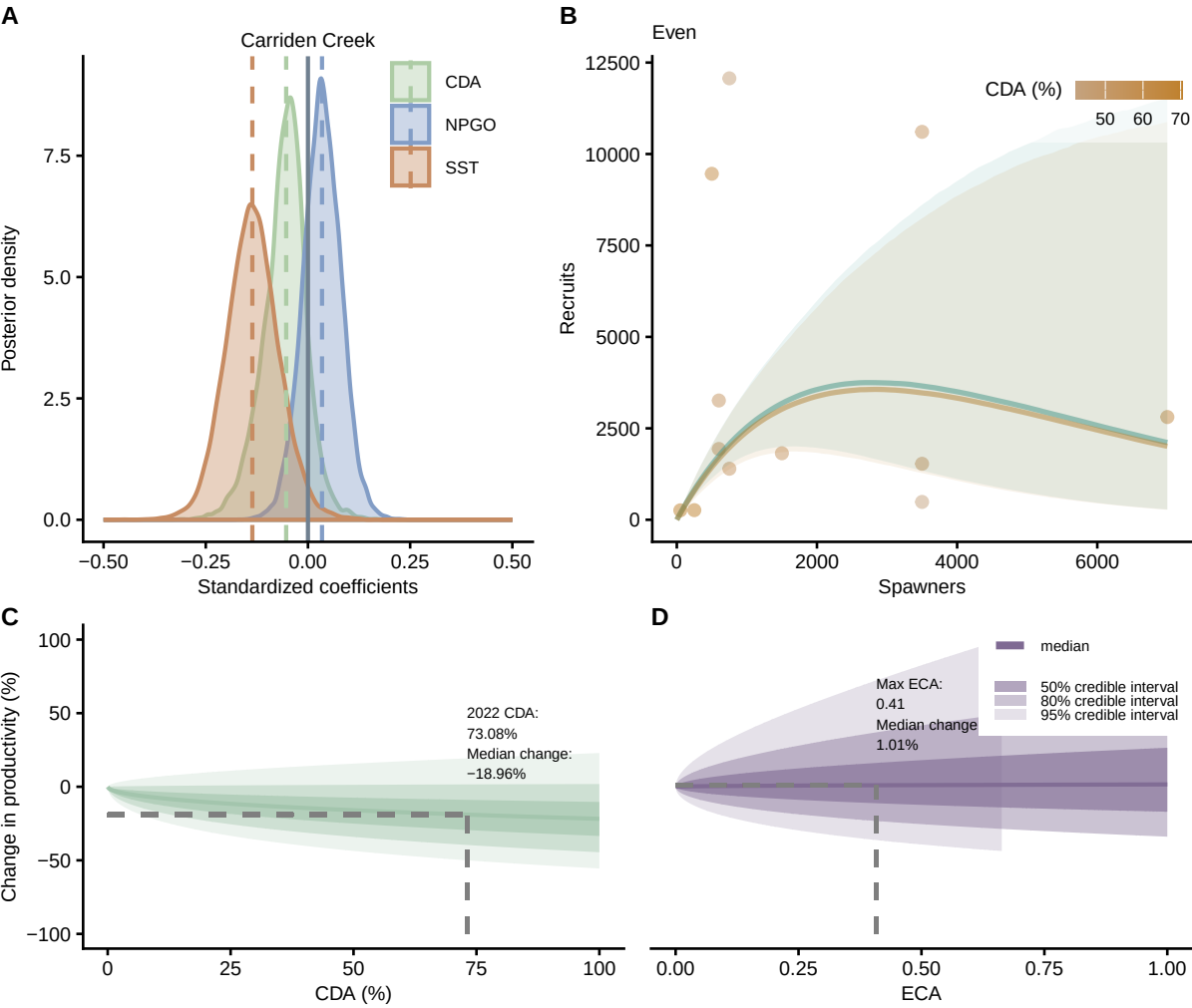


CARRIDEN CREEK

Data for Pink Salmon

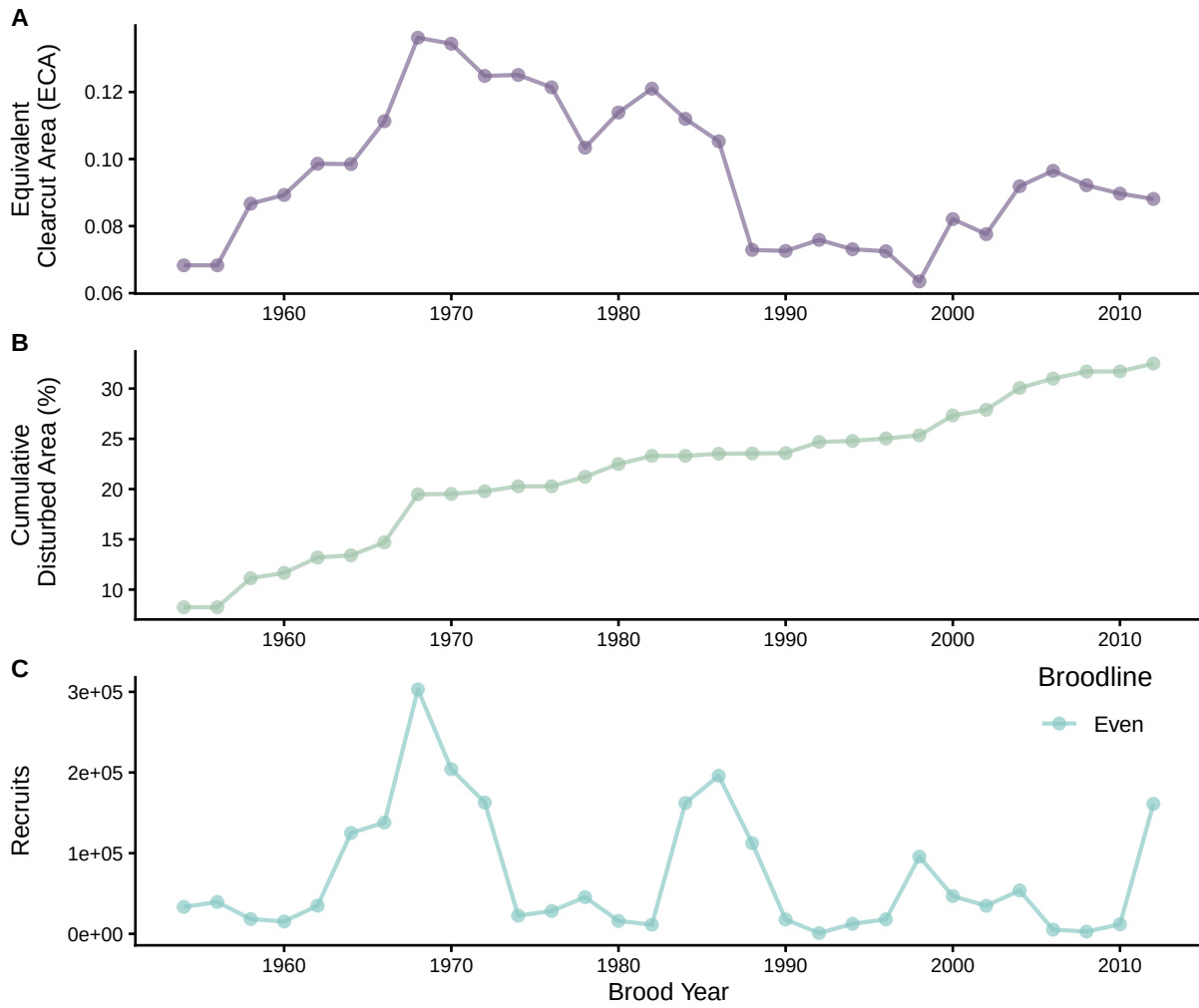


Model Results for Pink Salmon



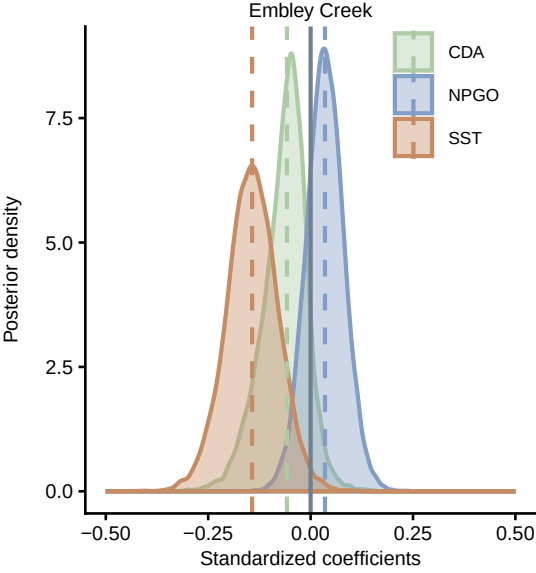
EMBLEY CREEK

Data for Pink Salmon

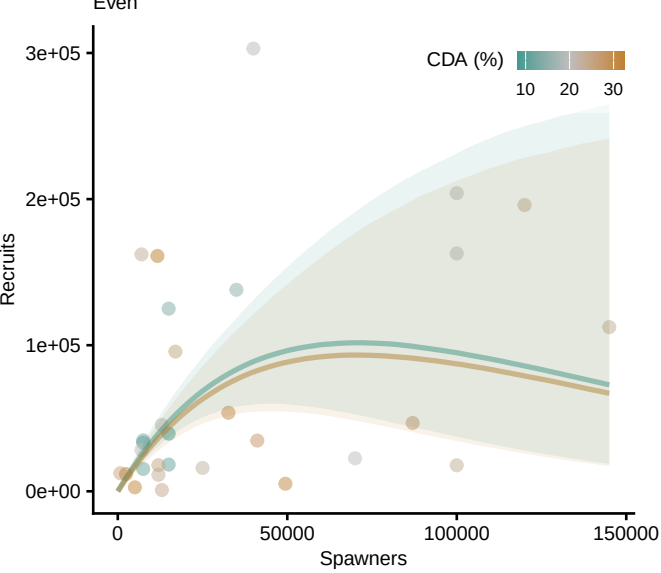


Model Results for Pink Salmon

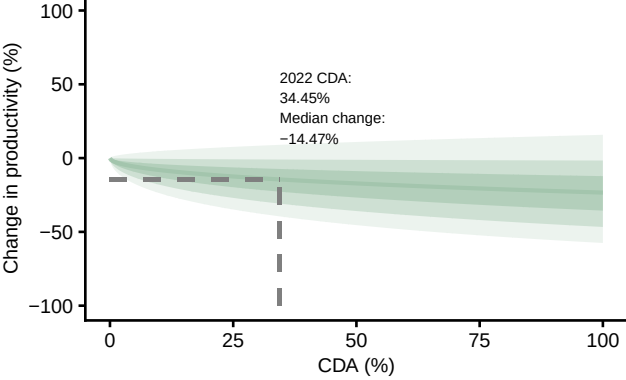
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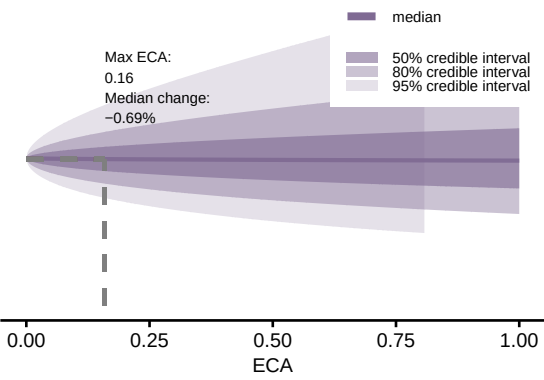
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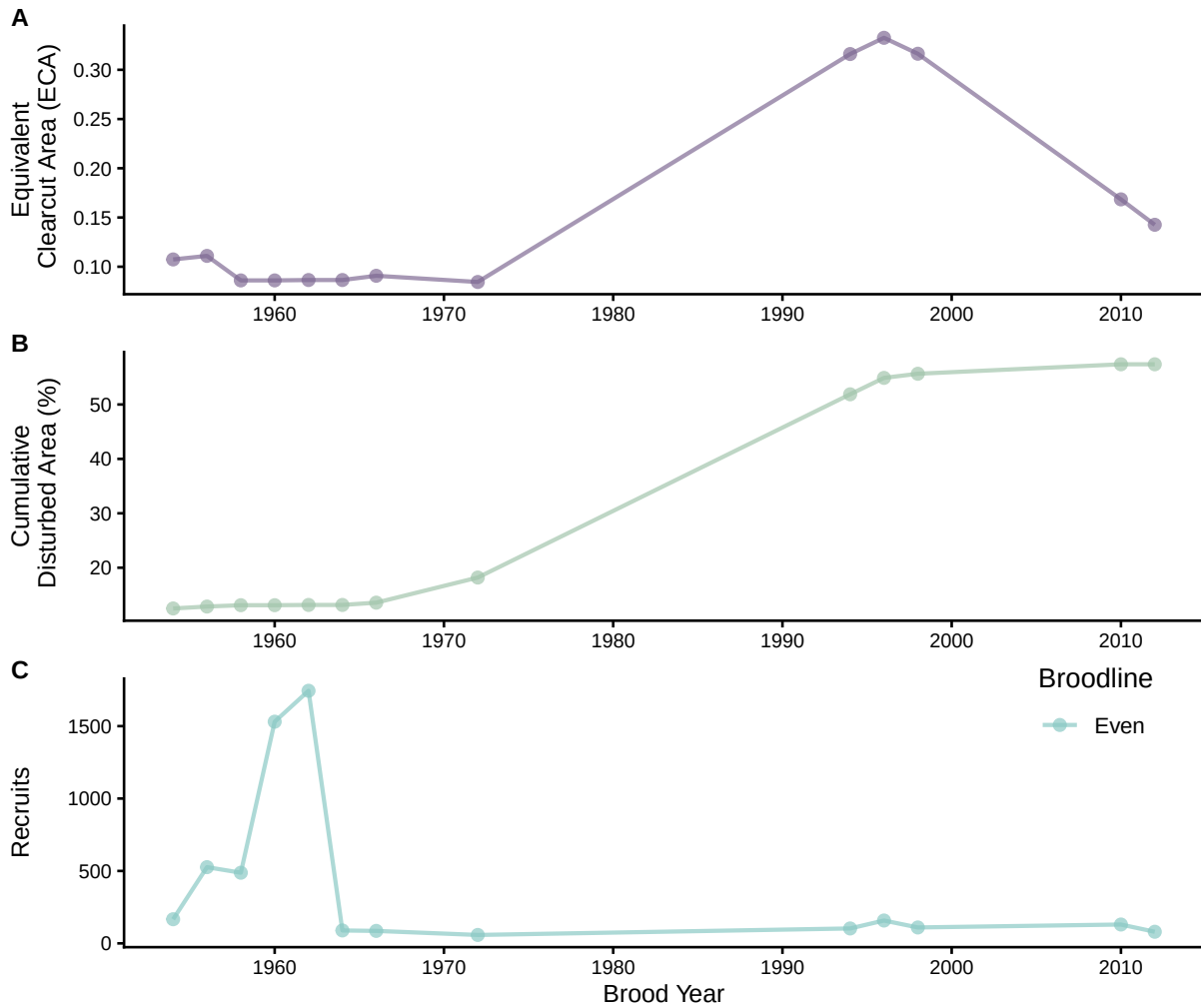


D

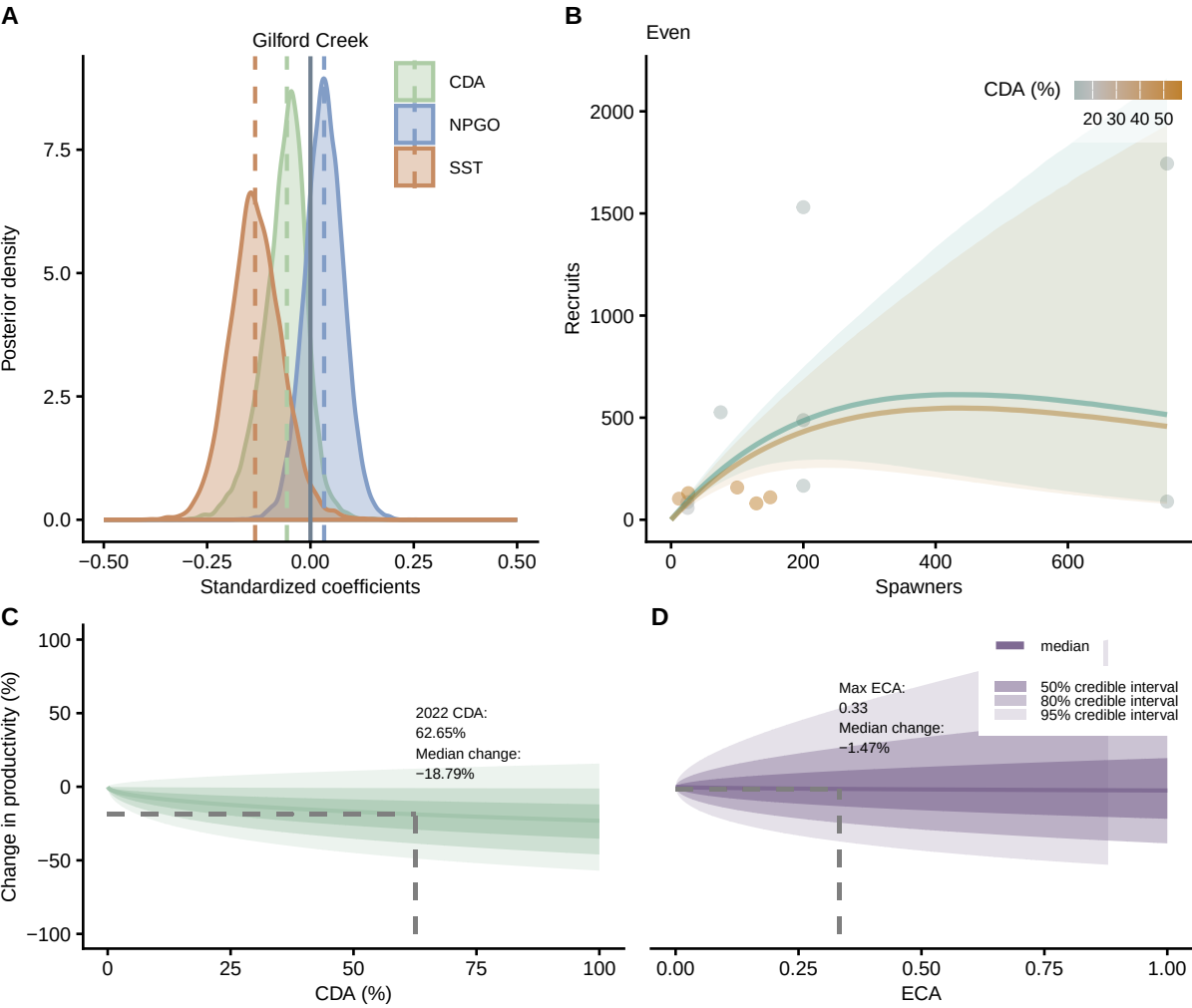


GILFORD CREEK

Data for Pink Salmon

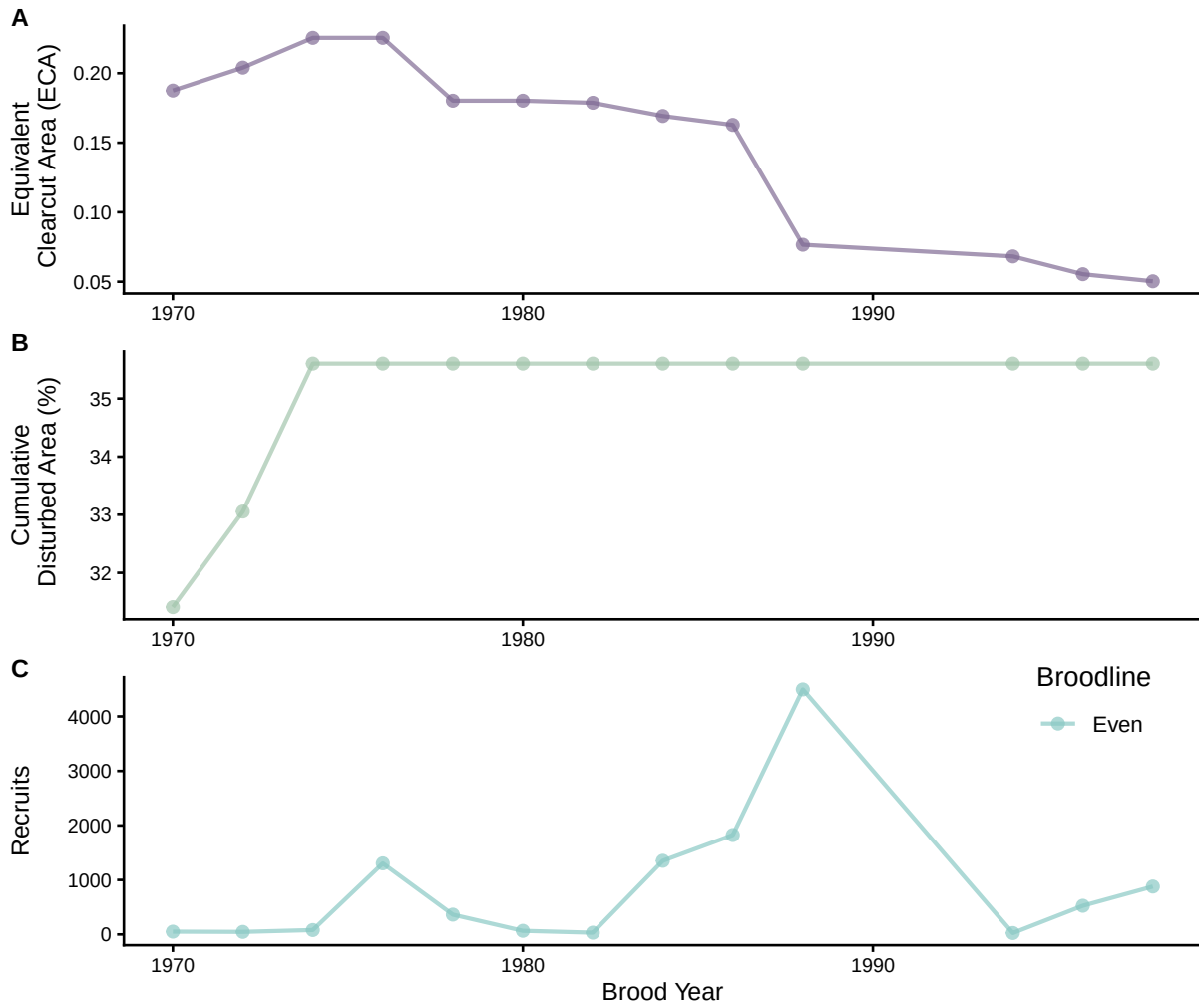


Model Results for Pink Salmon



MCALISTER CREEK

Data for Pink Salmon



Model Results for Pink Salmon

